

Overview of mathematical and statistical necessities for population genomics

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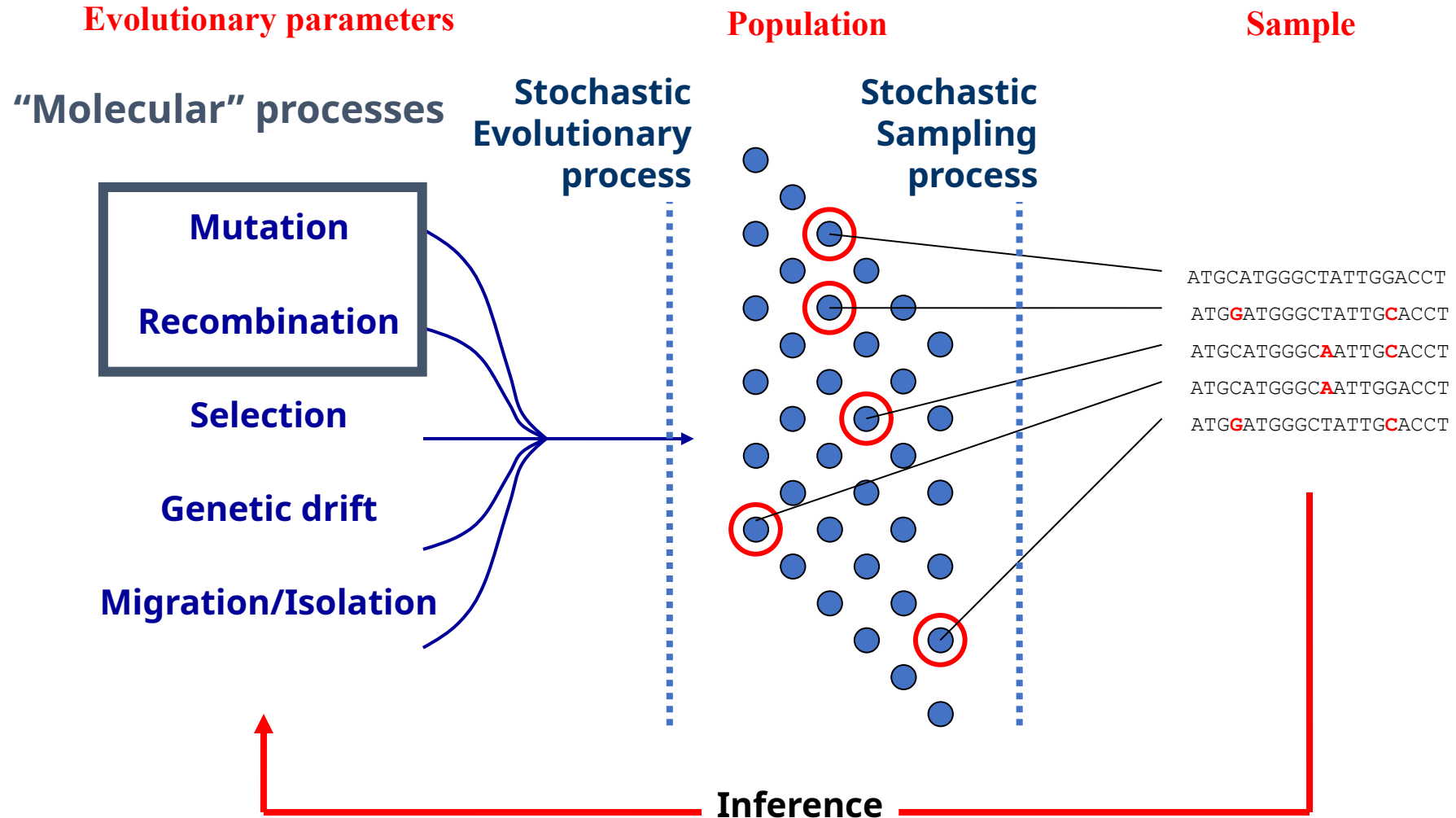
PRINCIPLES OF EVOLUTION

- 1) THERE IS AN ENVIRONMENT FOLLOWING ITS OWN RULES AND CONSTRAINTS (IN OUR CASE, REALITY BASED ON PHYSICAL CONSTRAINTS).
- 2) THERE ARE ELEMENTS THAT AUTOREPLICATE WITH ERRORS (IN OUR CASE DNA).
- 3) THE NUMBER OF ALLOWED ELEMENTS AT A GIVEN MOMENT IS LIMITED (IN OUR CASE, A FINITE POPULATION OR SPECIES).
- 4) THE ELEMENTS CONTAINS THE INFORMATION TO INTERACT WITH THE ENVIRONMENT (IN OUR CASE, OUR BODIES).

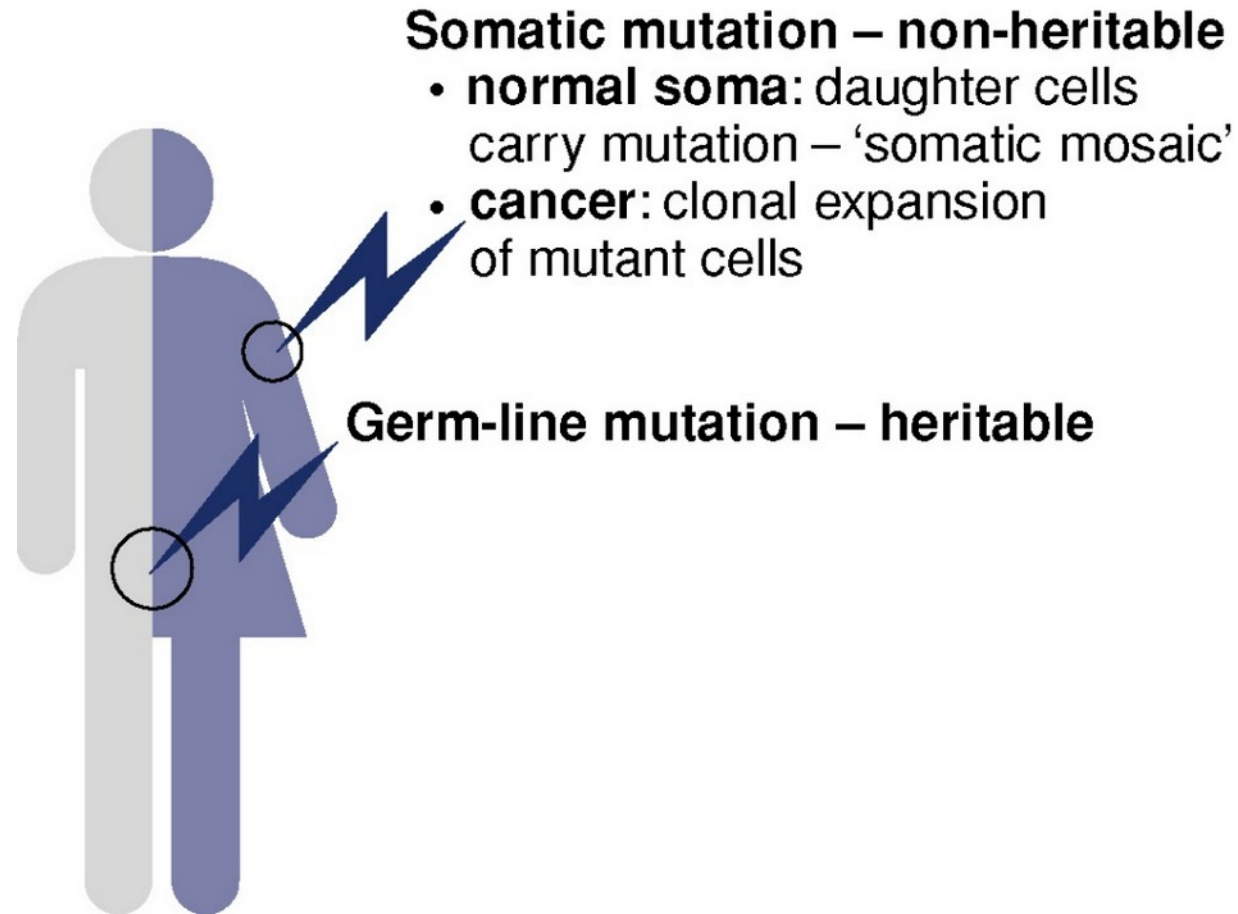
ANY SYSTEM THAT FOLLOWS THESE RULES WILL LOOK FOR OPTIMAL STATES



WHAT IS THE PURPOSE OF POPULATION GENETICS



HOW DOES EVOLUTION WORK?

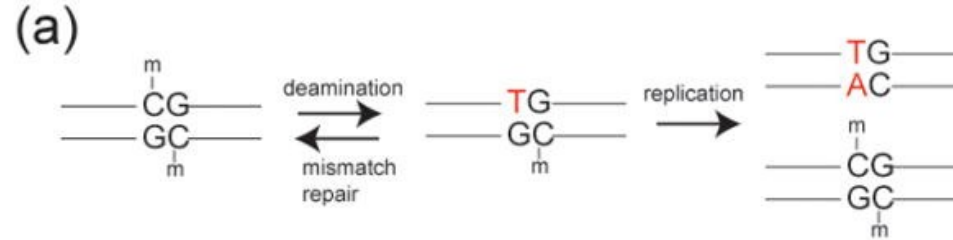


Human Evolutionary Genetics

Types of (observed) mutations

Trends Genet. 2013 Oct; 29(10): 575–584

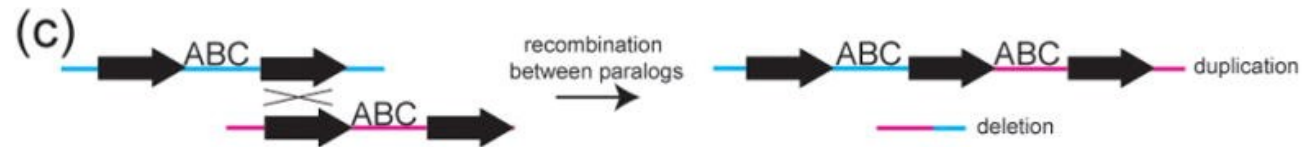
CpG & Single nucleotide variants



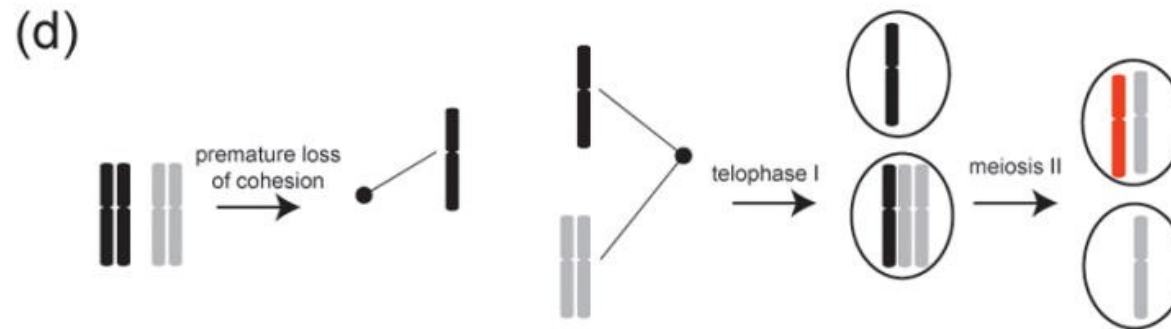
Repetitions & Indels



Large Duplications/Deletions & paralogs

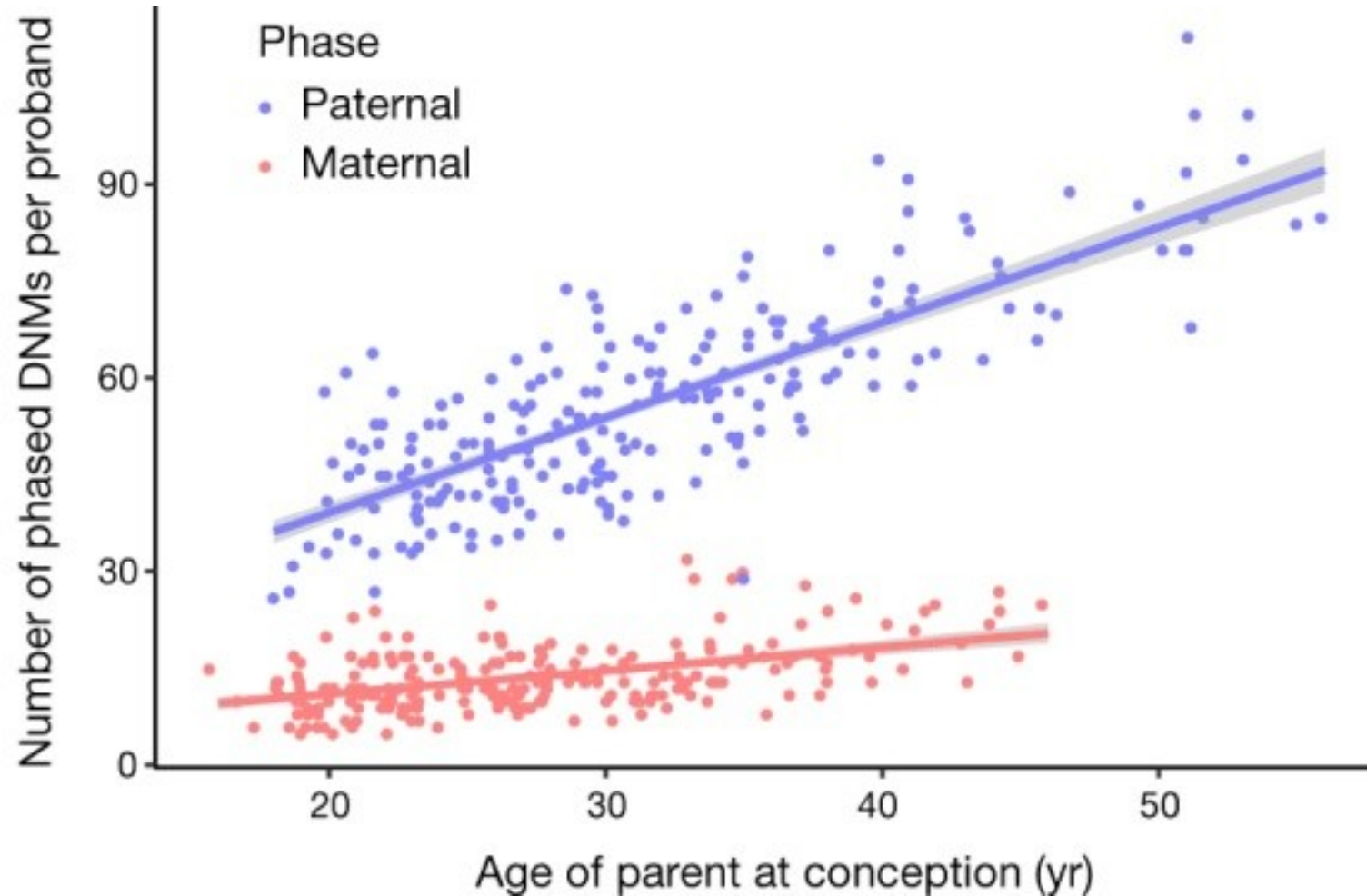


Trisomies

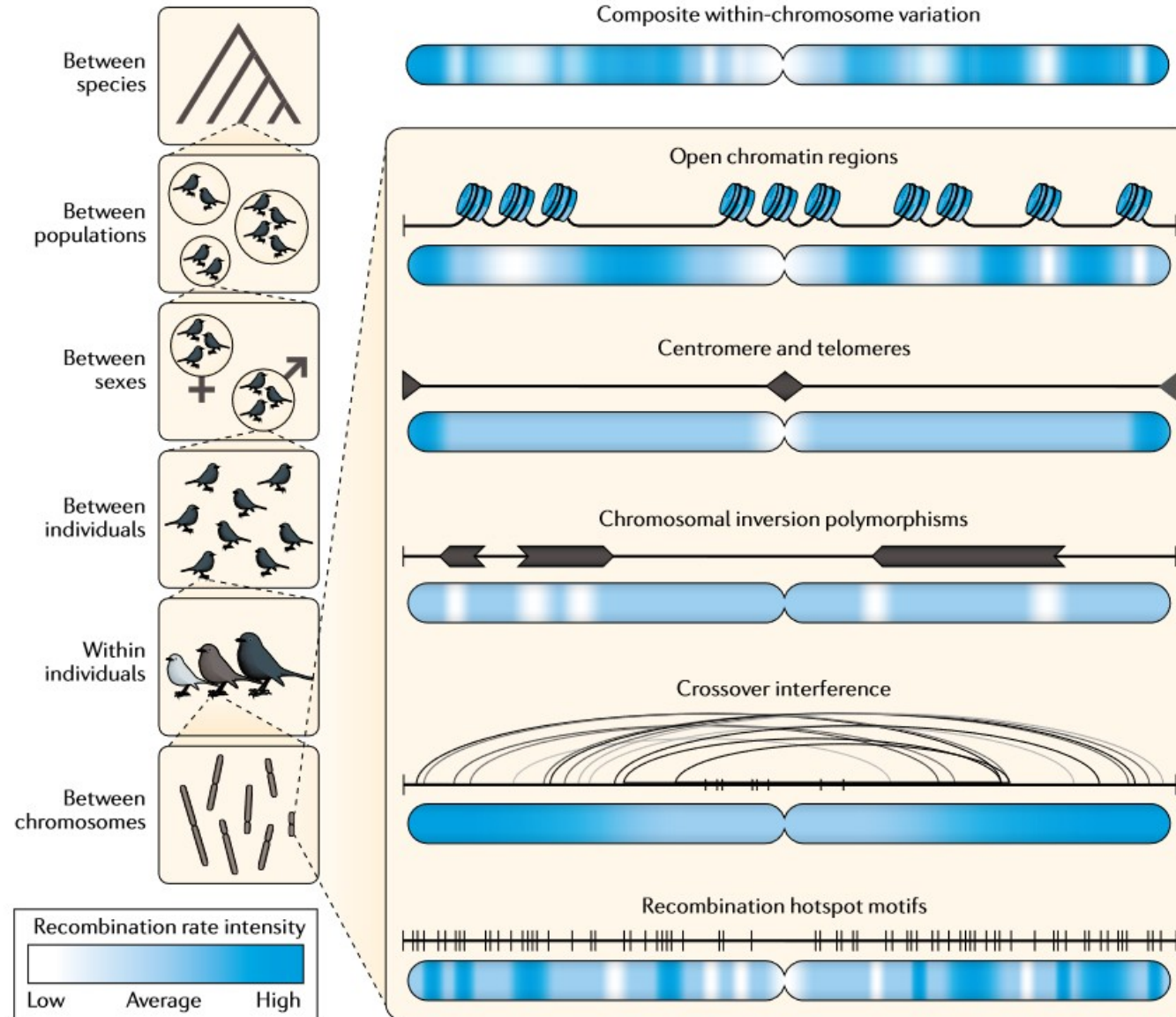


Mutation rates depend on age and sex

H Jónsson *et al. Nature* 1–4 (2017) doi:10.1038/nature24018

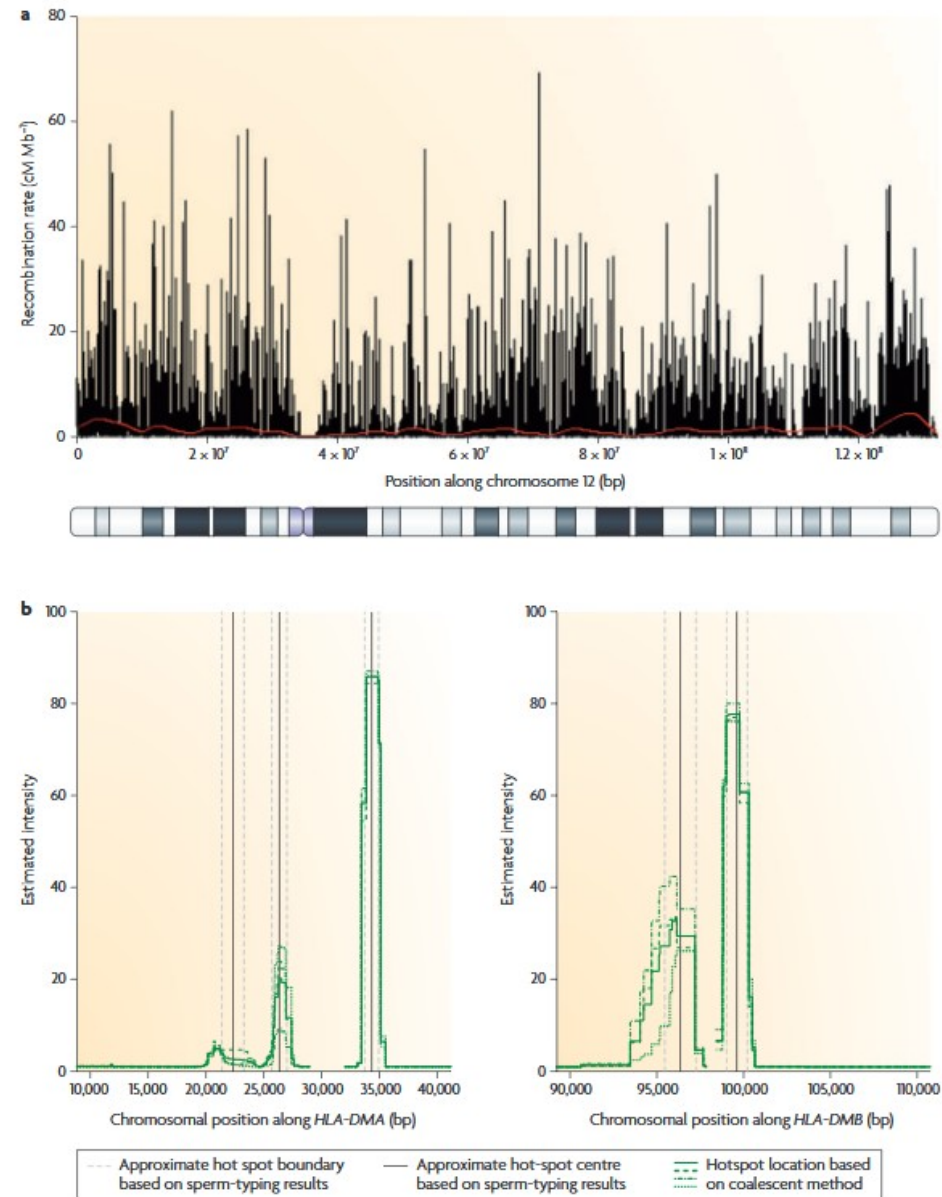


Recombination

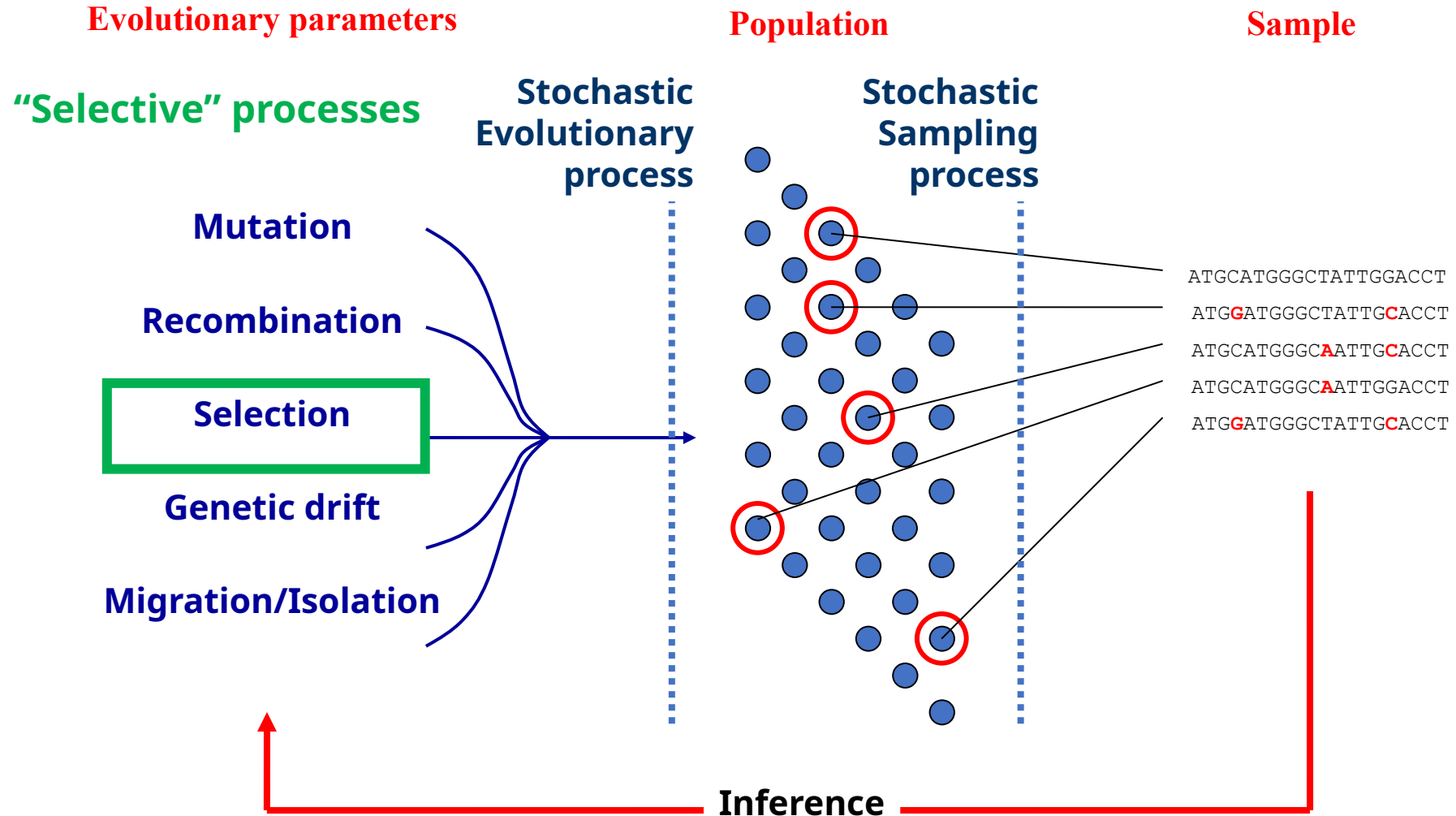


Recombination

Nature Reviews Genetics volume 11, pages 221–233 (2010)



WHAT IS THE PURPOSE OF POPULATION GENETICS



HOW DOES EVOLUTION WORK?

REQUIRES THAT REPLICATES COMPETE AGAINST EACH OTHER FOR THE
SAME ENVIRONMENT

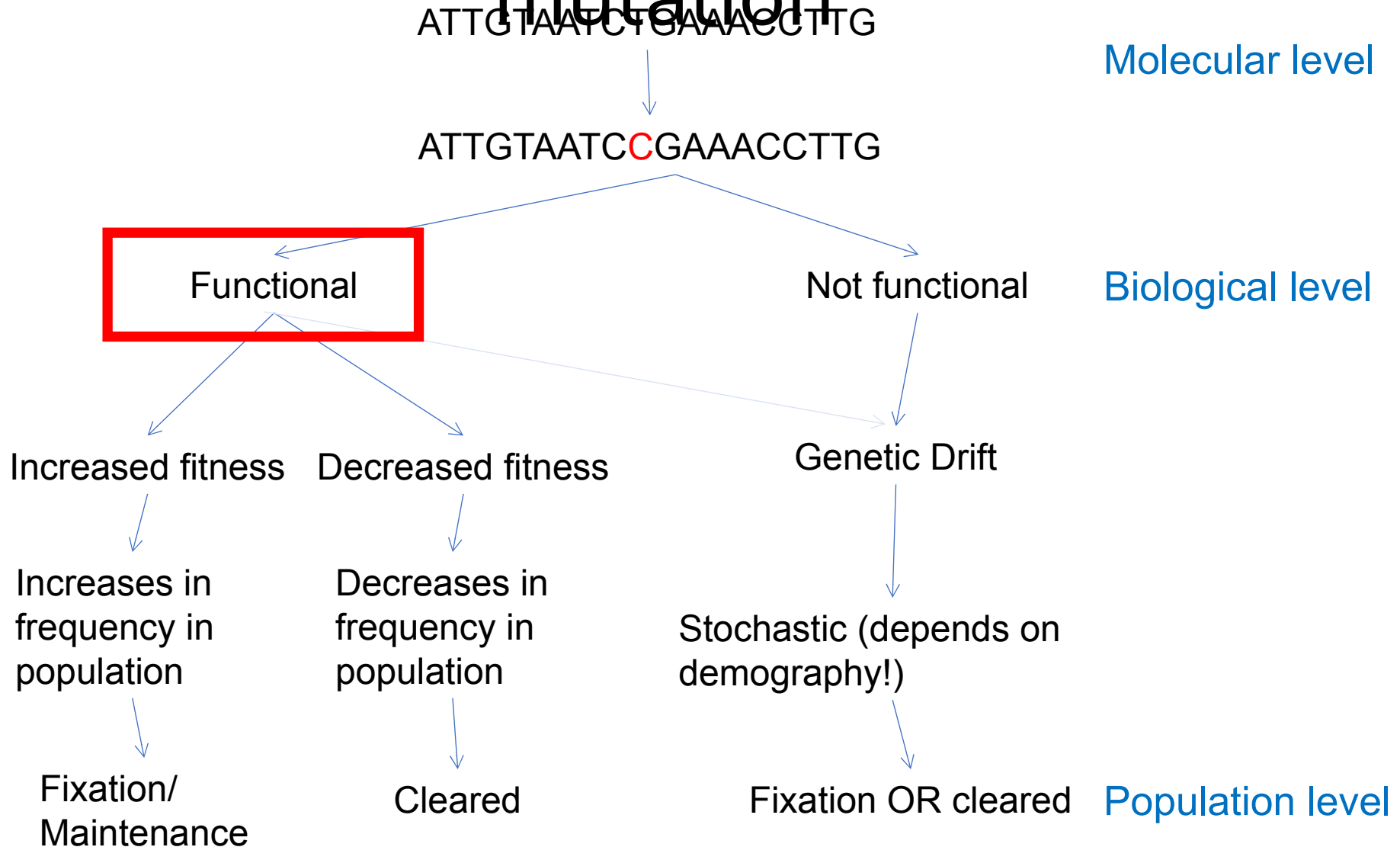


SELECTION: WORKS BY COMPARISON!



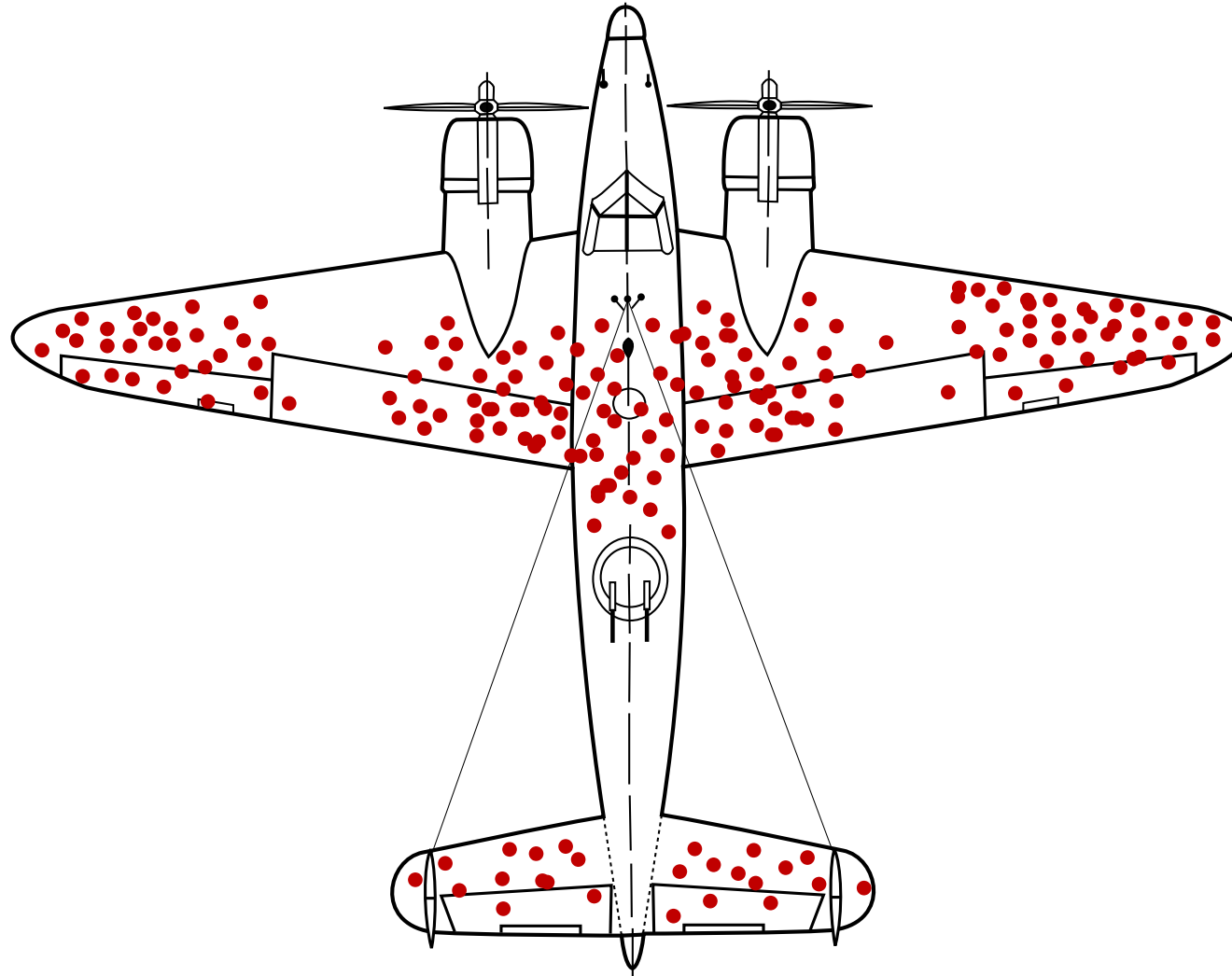
John Lydgate (1440): *Odyous of olde been comparisonis, And of comparisonis engendyrd is haterede. (Comparisons are odious)*

The evolutionary fate of a mutation

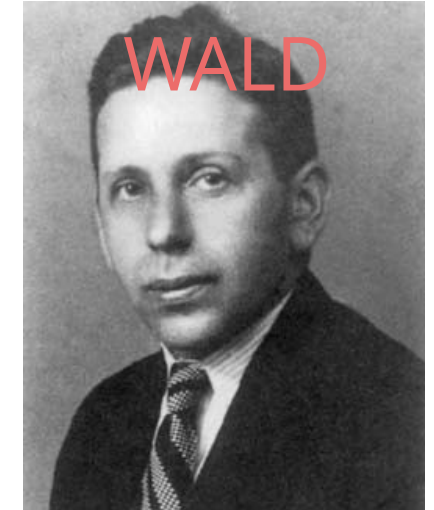


MUTATIONS AND BULLETS

By Martin Grandjean (vector), McGeddon (picture), Cameron Moll
(concept) - Own work, CC BY-SA 4.0,
<https://commons.wikimedia.org/w/index.php?curid=102017718>



ABRAHAM
WALD



By
<http://www-history.mcs.st-andrews.ac.uk/PictDisplay/Wald.html>, PD-US,
<https://en.wikipedia.org/w/index.php?curid=15142725>

MUTATIONS AND BULLETS

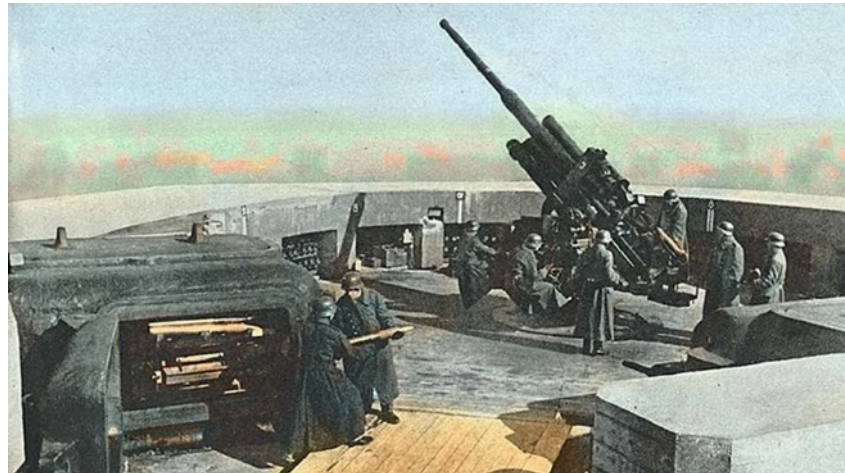
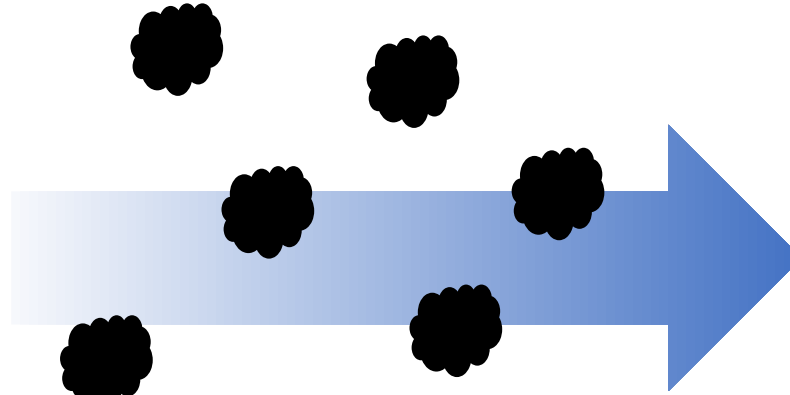
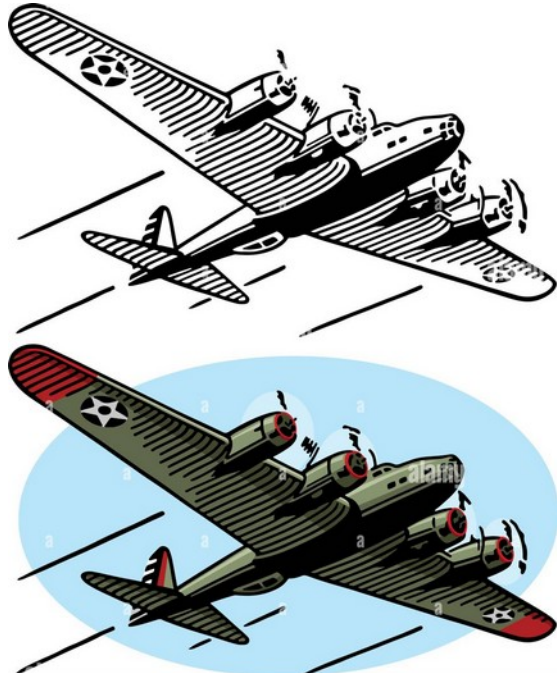
Abraham Wald and the Missing Bullet Holes

Section of plane	Bullet holes per square foot
Engine	1.11
Fuselage	1.73
Fuel system	1.55
Rest of the plane	1.8

The officers saw an opportunity for efficiency; you can get the same protection with less armor if you concentrate the armor on the places with the greatest need, where the planes are getting hit the most. But exactly how much more armor belonged on those parts of the plane? That was the answer they came to Wald for. It wasn't the answer they got.

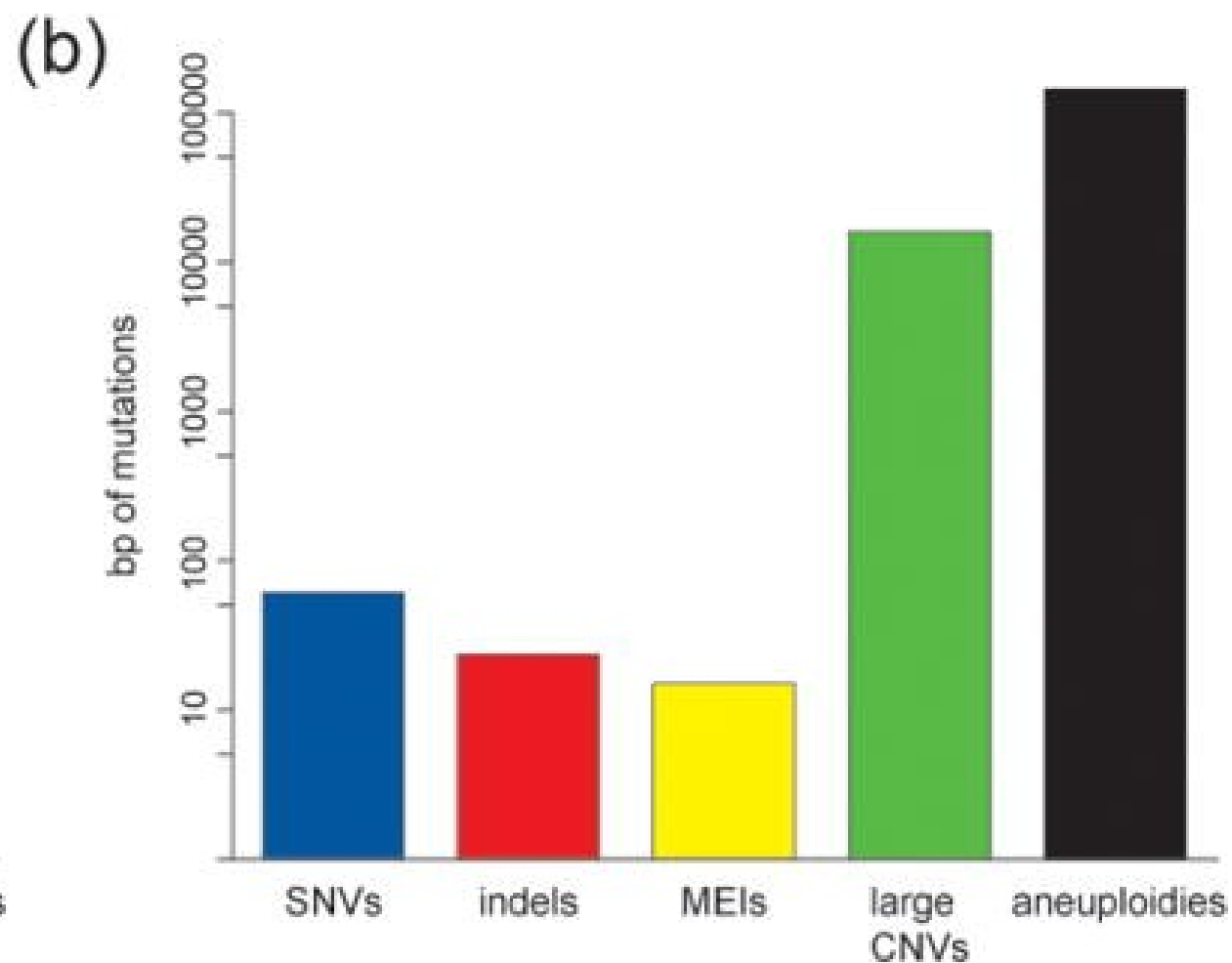
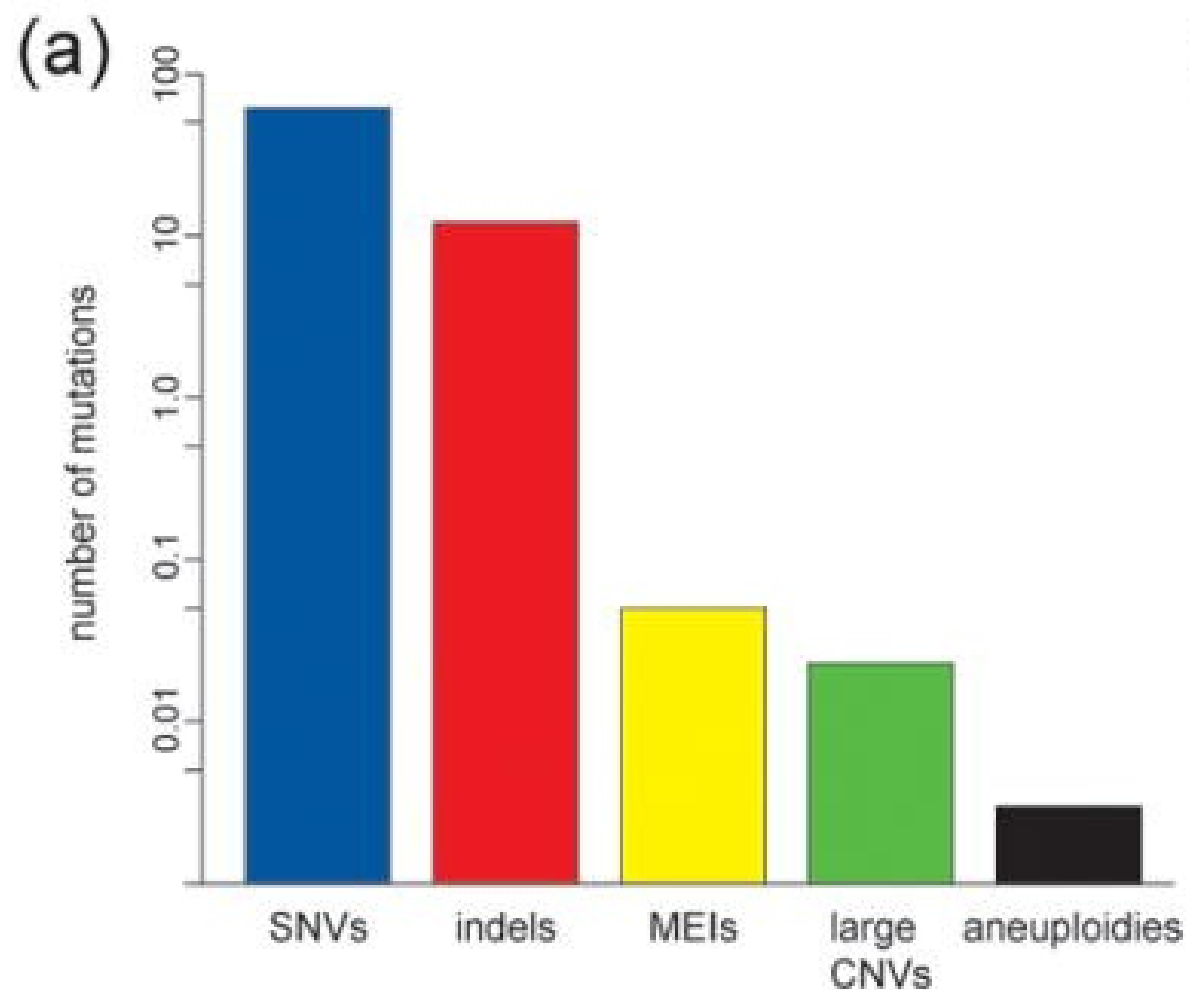
MUTATIONS AND BULLETS

Abraham Wald and the Missing Bullet Holes



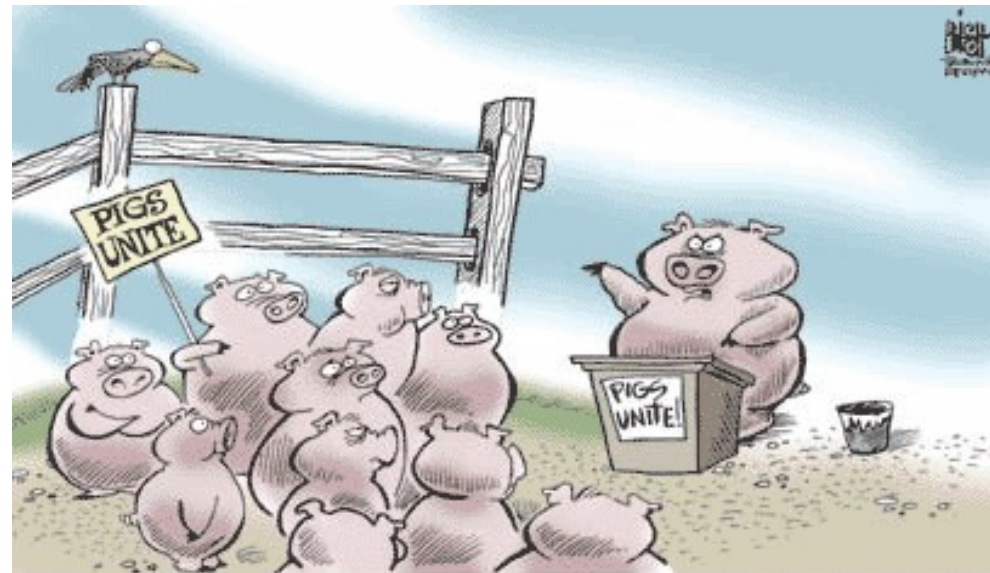
Types of (observed) mutations

Trends Genet. 2013 Oct; 29(10): 575–584



Some reasons for genetic adaptation

- Adaptation to new environments
- Food production – new diets
- Population increase – new diseases



THE TURKEYS HAVE BIRD FLU. THE COWS HAVE MAD COW DISEASE. I'M TELLING YOU, BOYS... UNLESS WE WANT TO SEE MORE HAM SERVED ON THANKSGIVING, WE'RE GOING TO HAVE TO GET OUR OWN DISEASE!

A CLASSICAL EXAMPLE: ADULT LACTOSE TOLERANCE

- Humans are mammals
- This means that, among other things, **we consume milk when we are babies**
- This is done thanks to the enzyme LACTASE (*LCT*)
- **Milk** is a complete source of energy and proteins + defense

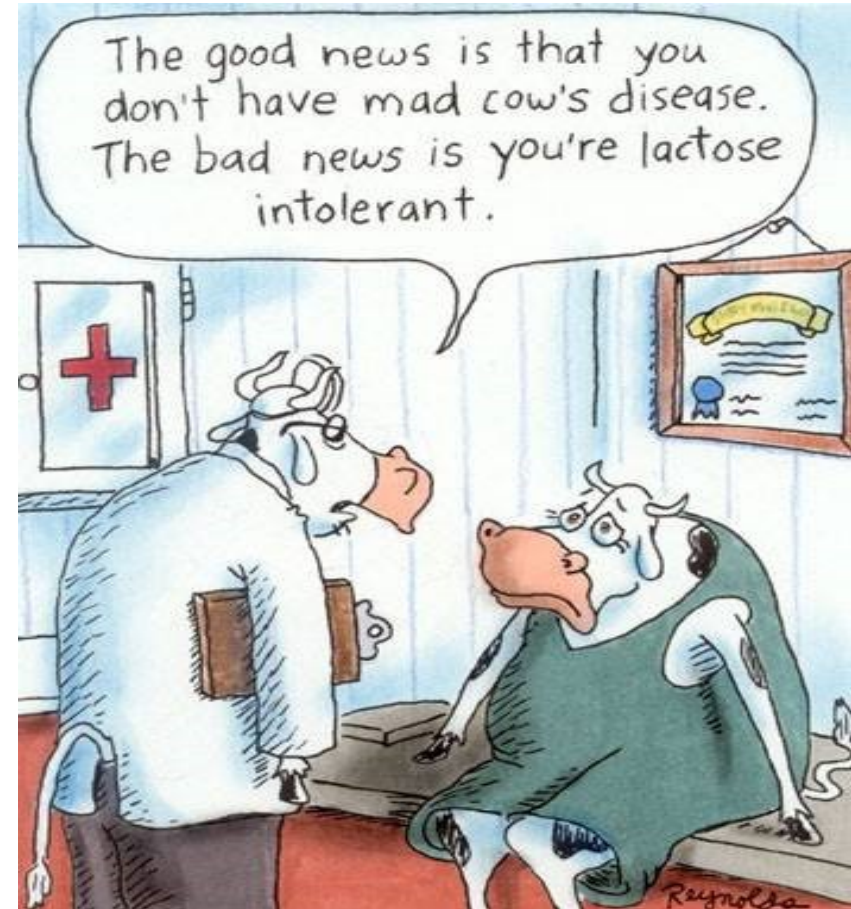
A CLASSICAL EXAMPLE: ADULT LACTOSE TOLERANCE



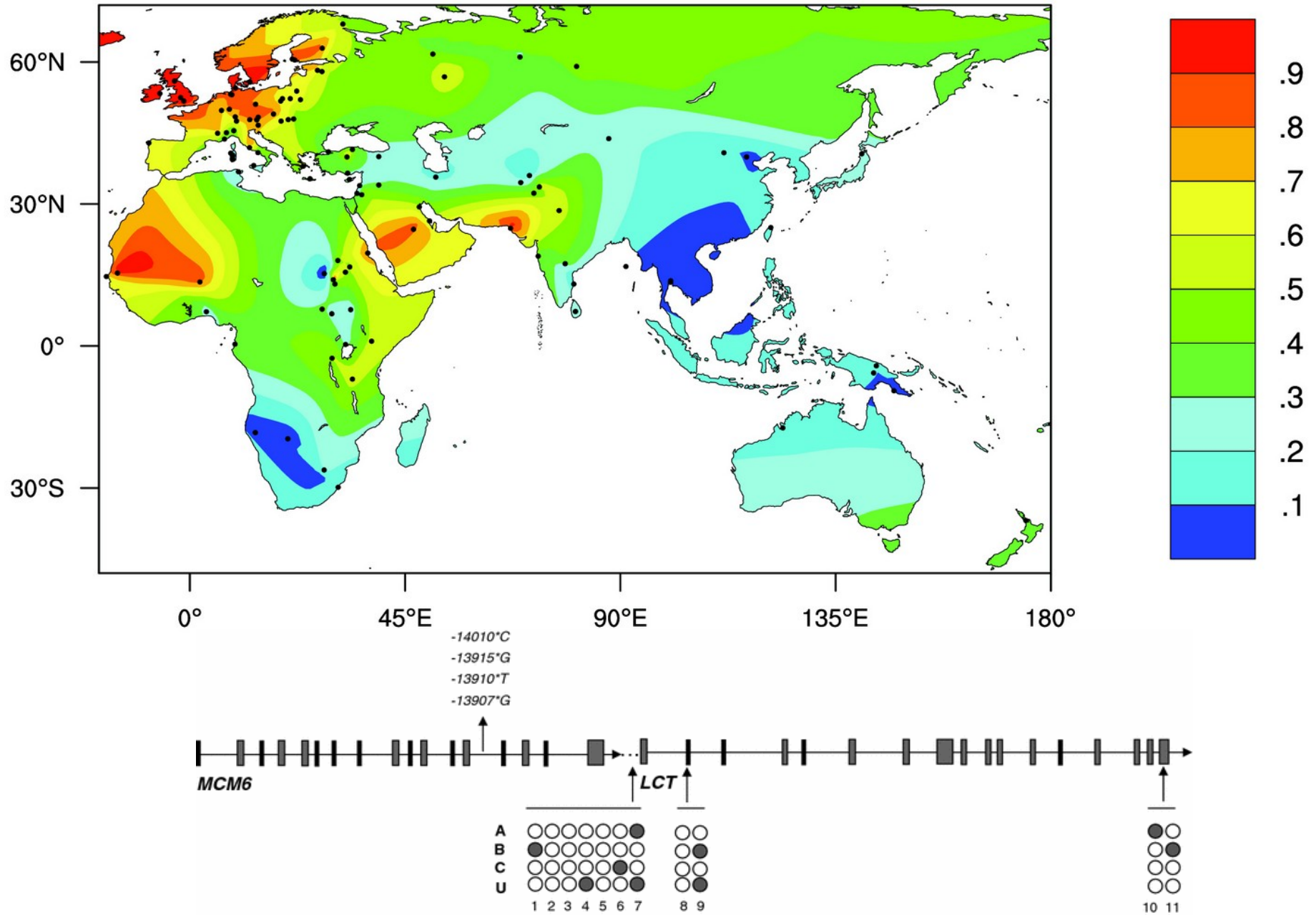
- The capacity to metabolize lactose **disappears** at adult age in almost all the mammal species
- *LCT* gene is not expressed anymore because there is no more maternal milk to drink
- Or is it not?

A CLASSICAL EXAMPLE: ADULT LACTOSE TOLERANCE

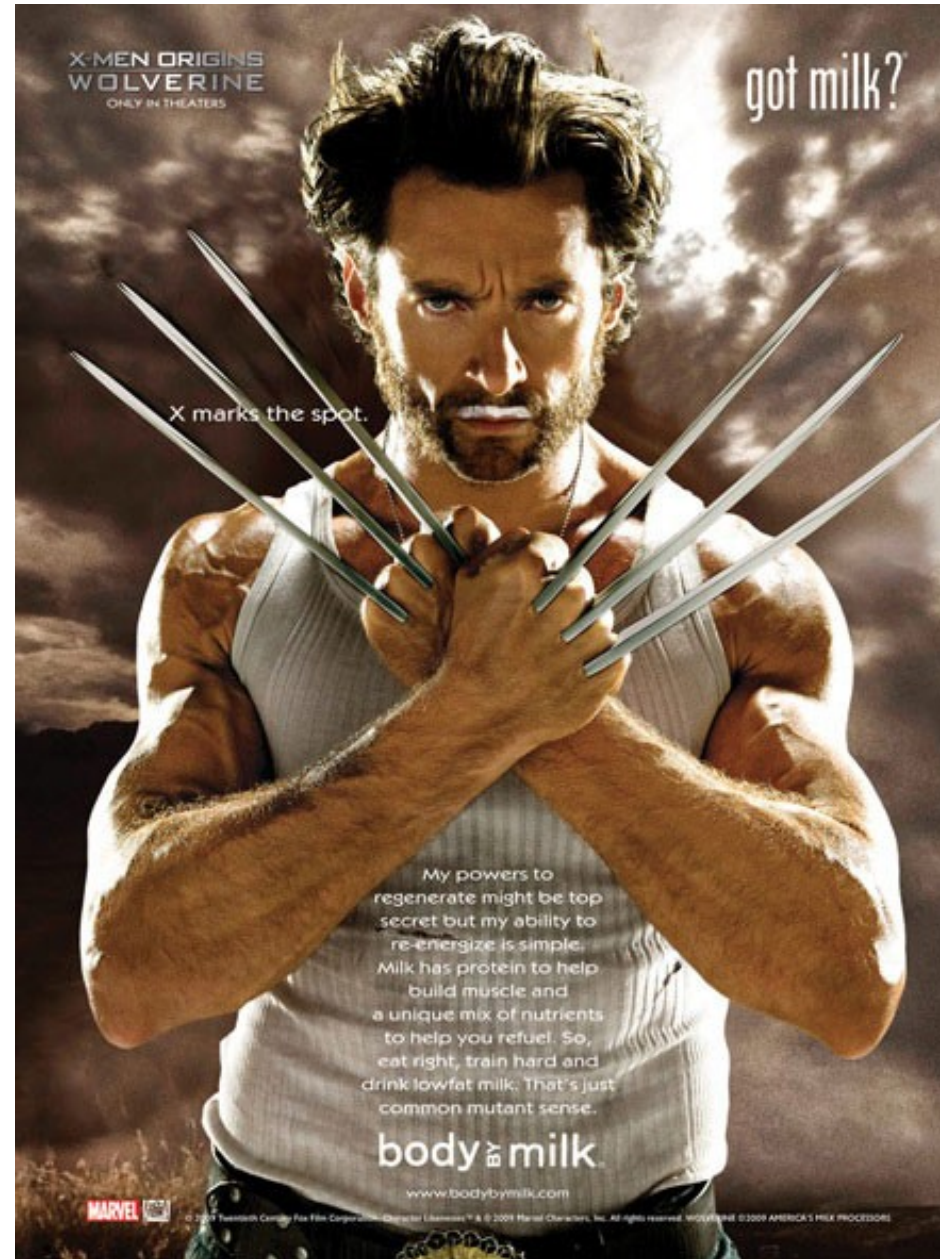
- How to recognize if you are **lactose intolerant**?
 - Can be asymptomatic
 - Gas production
 - Diarrhea
 - Related to diseases such as inflammatory bowel disease
 - Usually lactose intolerant people “don’t like milk”



A CLASSICAL EXAMPLE: ADULT LACTOSE TOLERANCE

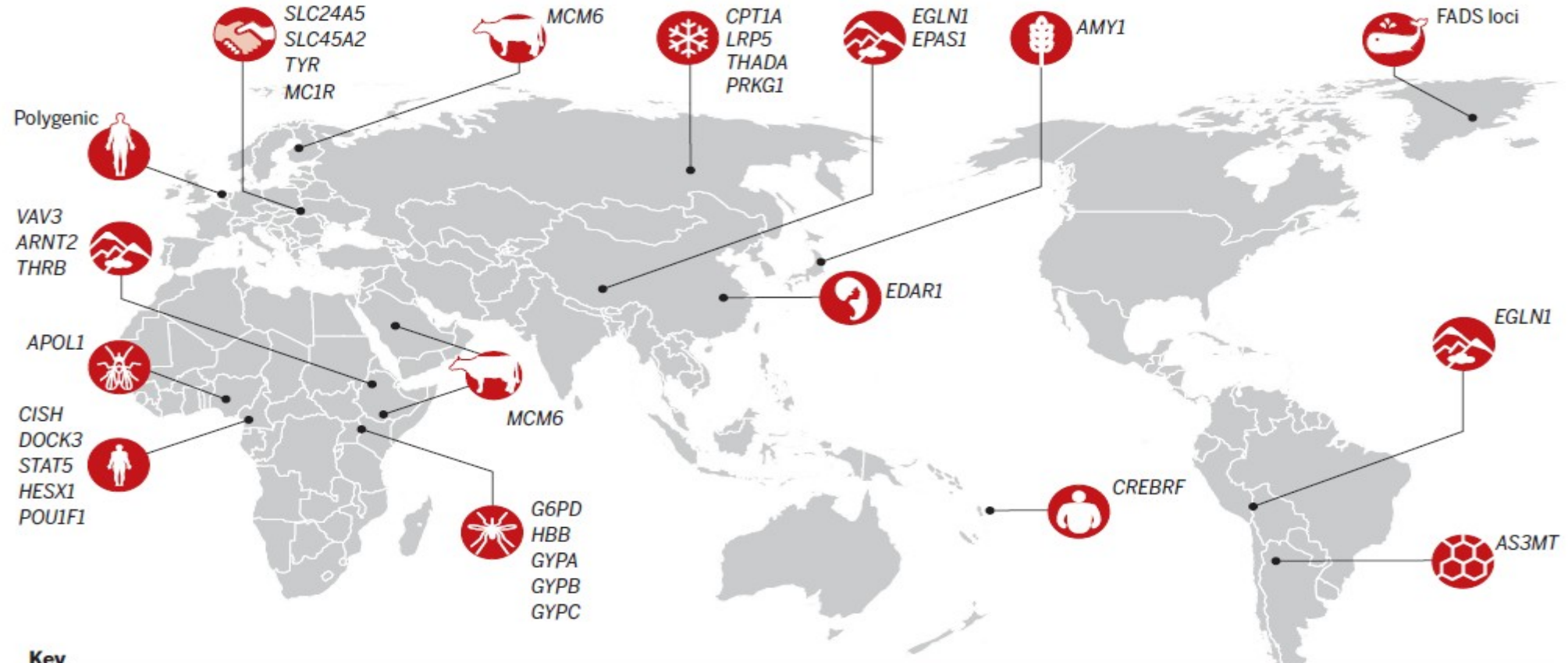


HOW A MUTANT LOOKS LIKE IN REALITY

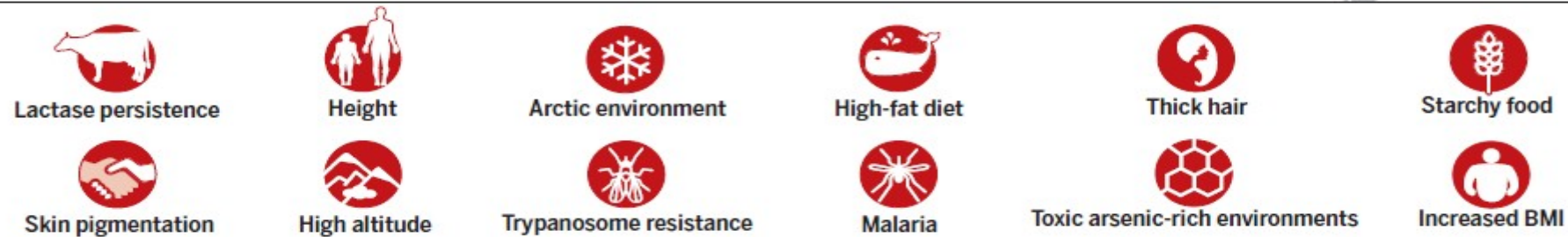


NOT ONLY LACTOSE TOLERANCE

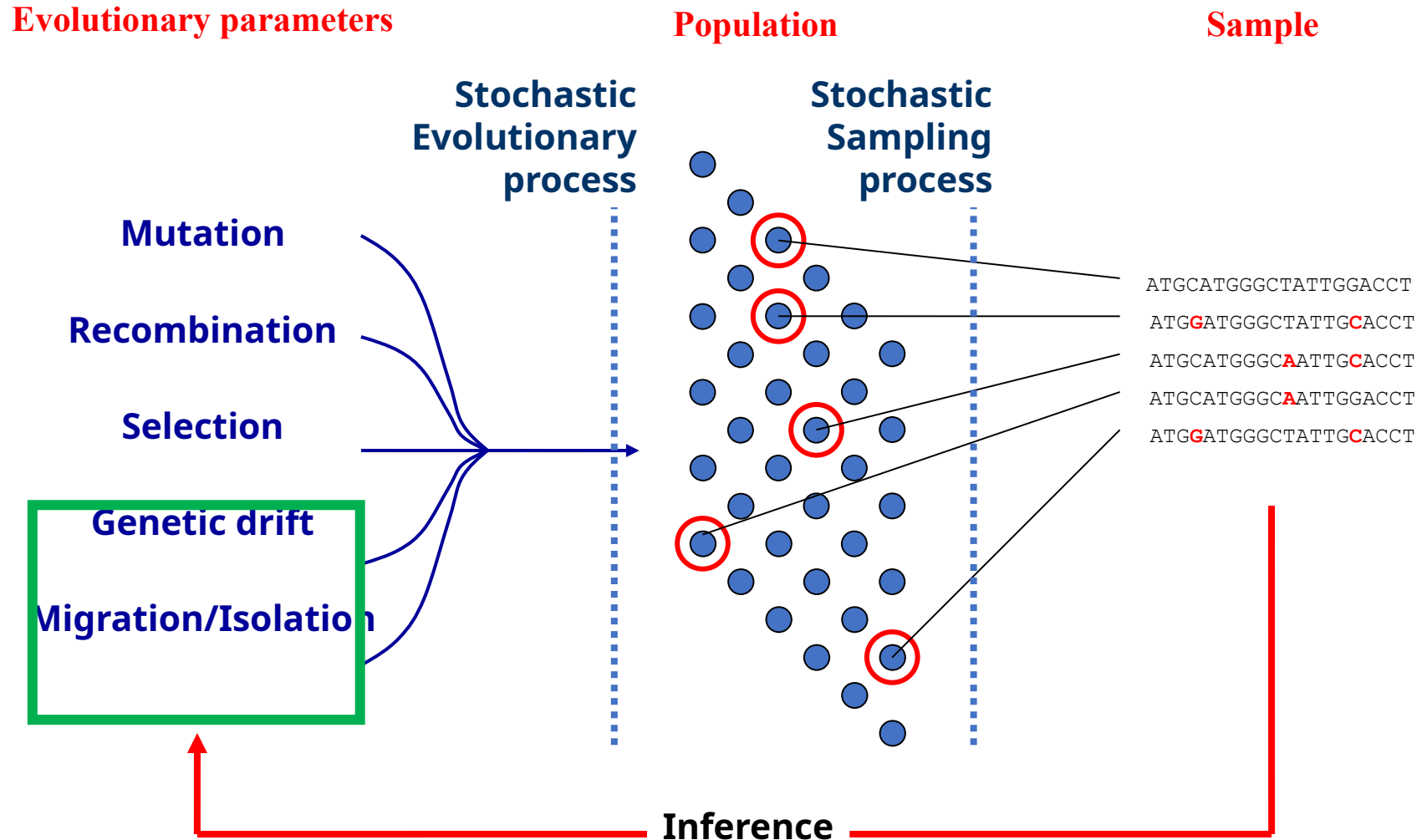
Fan et al Science 2016



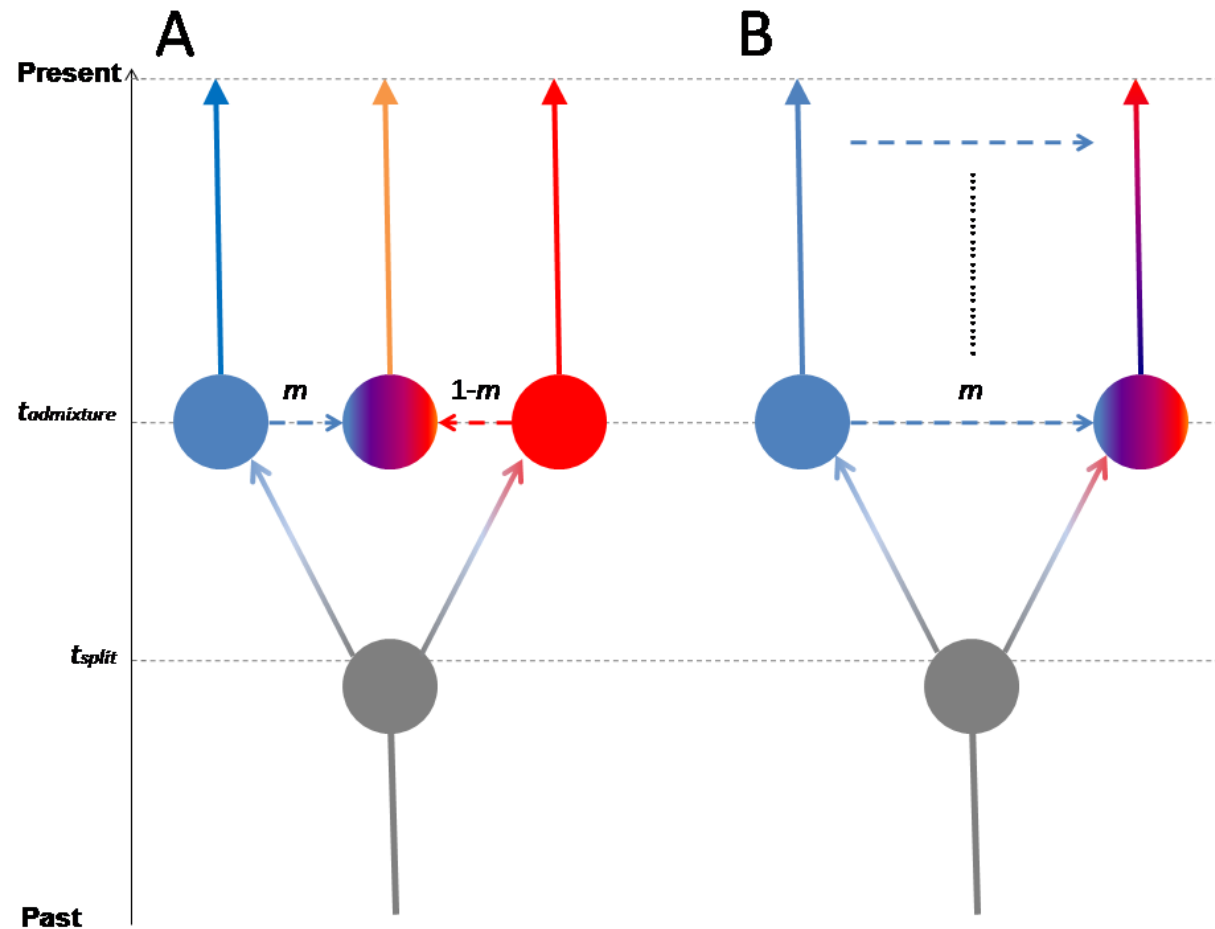
Key



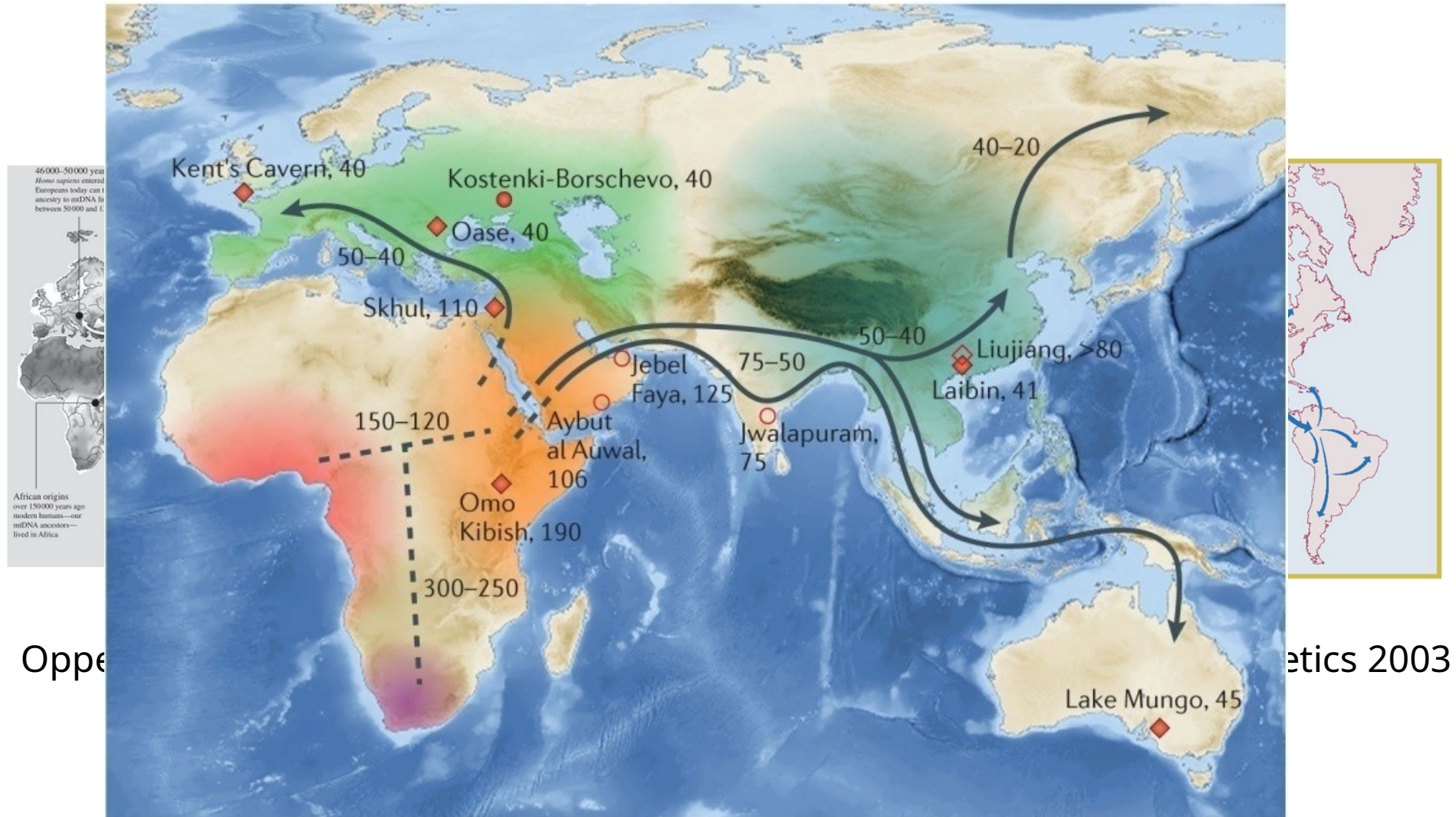
WHAT IS THE PURPOSE OF POPULATION GENETICS



Why is there population substructure?



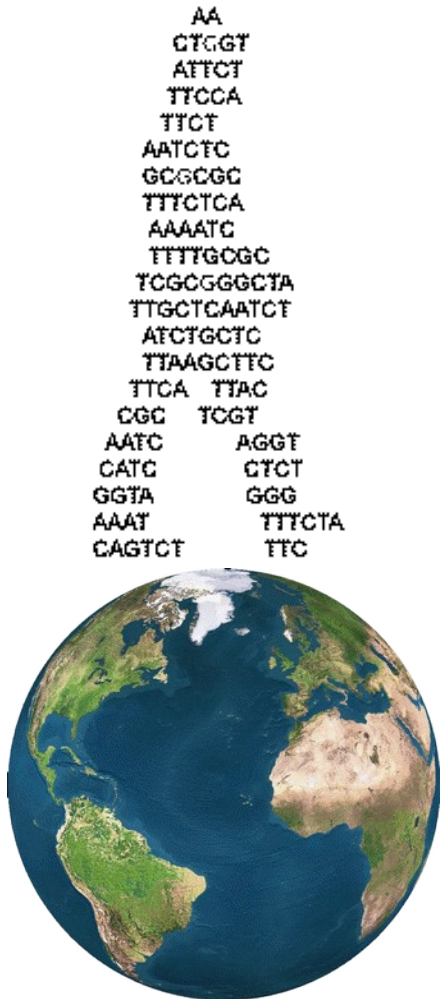
Why is there population substructure?



Oppen

etics 2003

Why is there population substructure?



- **Historical** factors
- **Physical** dependence of mating
- **Geographic** distribution of cultural factors affecting mating
 - Religion
 - Language
 - ...
- “Everything is related to everything else, but near things are more related than distant things” (Tobler's first law of geography)

WHAT IS THE PURPOSE OF POPULATION GENETICS

Applications

Evolutionary parameters

Population

Sample

Conservation
genetics

Evolutionary
medicine

(PUT YOUR
OWNS)

“Molecular” processes

Stochastic
Evolutionary
process

Stochastic
Sampling
process

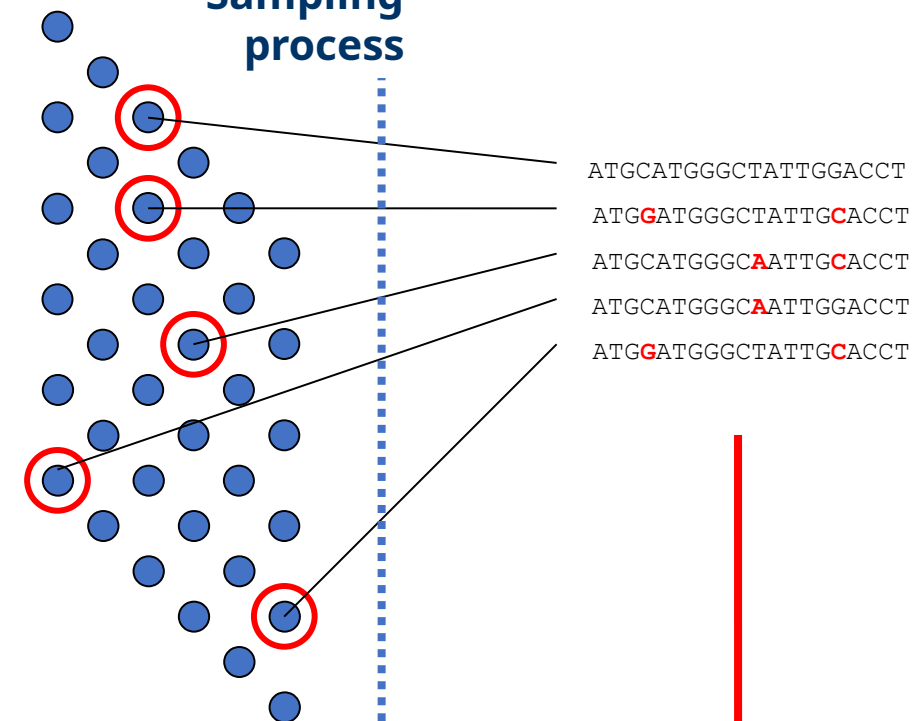
Mutation

Recombination

Selection

Genetic drift

Migration/Isolation

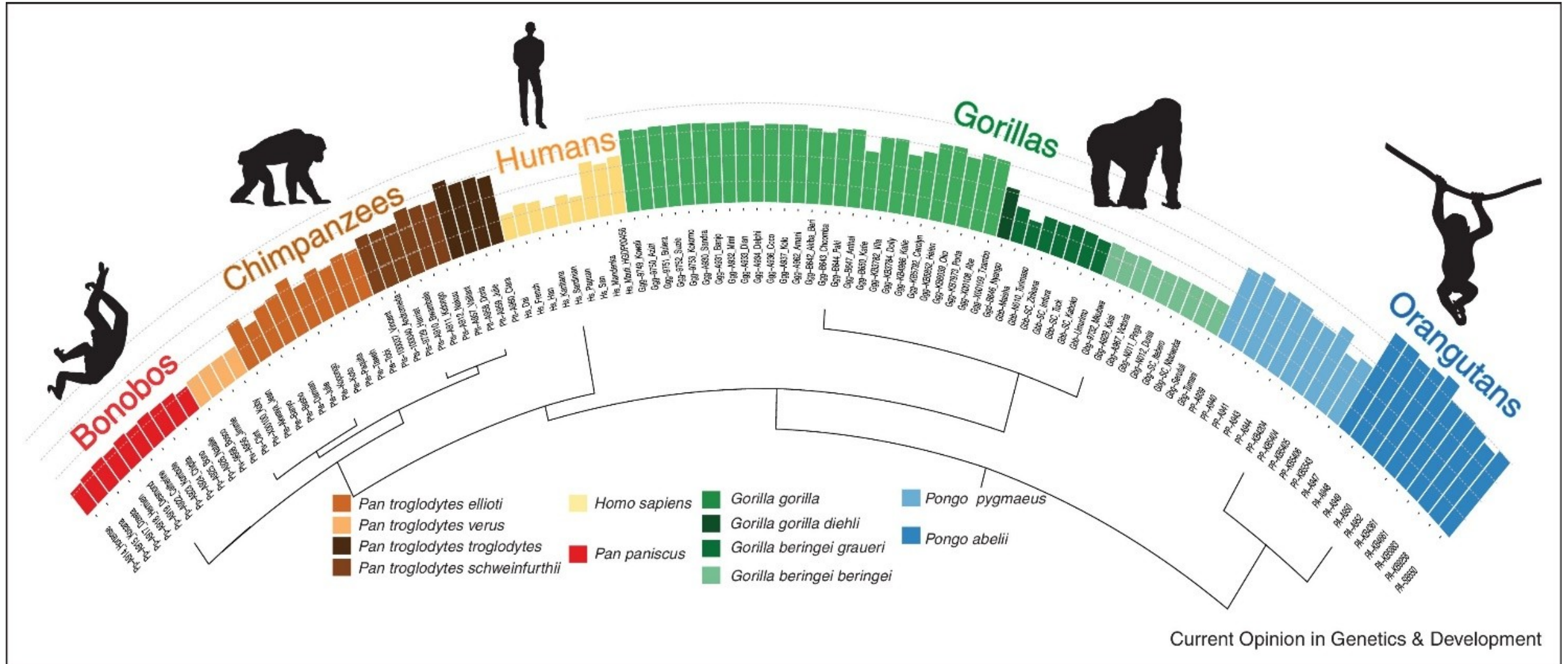


Inference

Adapted from McVean

GENETIC DIVERSITY AND DEMOGRAPHIC HISTORY?

Current Opinion in Genetics & Development Volume 41, December 2016, Pages 124-129



CONSERVATION GENETICS AND GENETIC DIVERSITY?

pnas 2021 118 (10) e2015096118

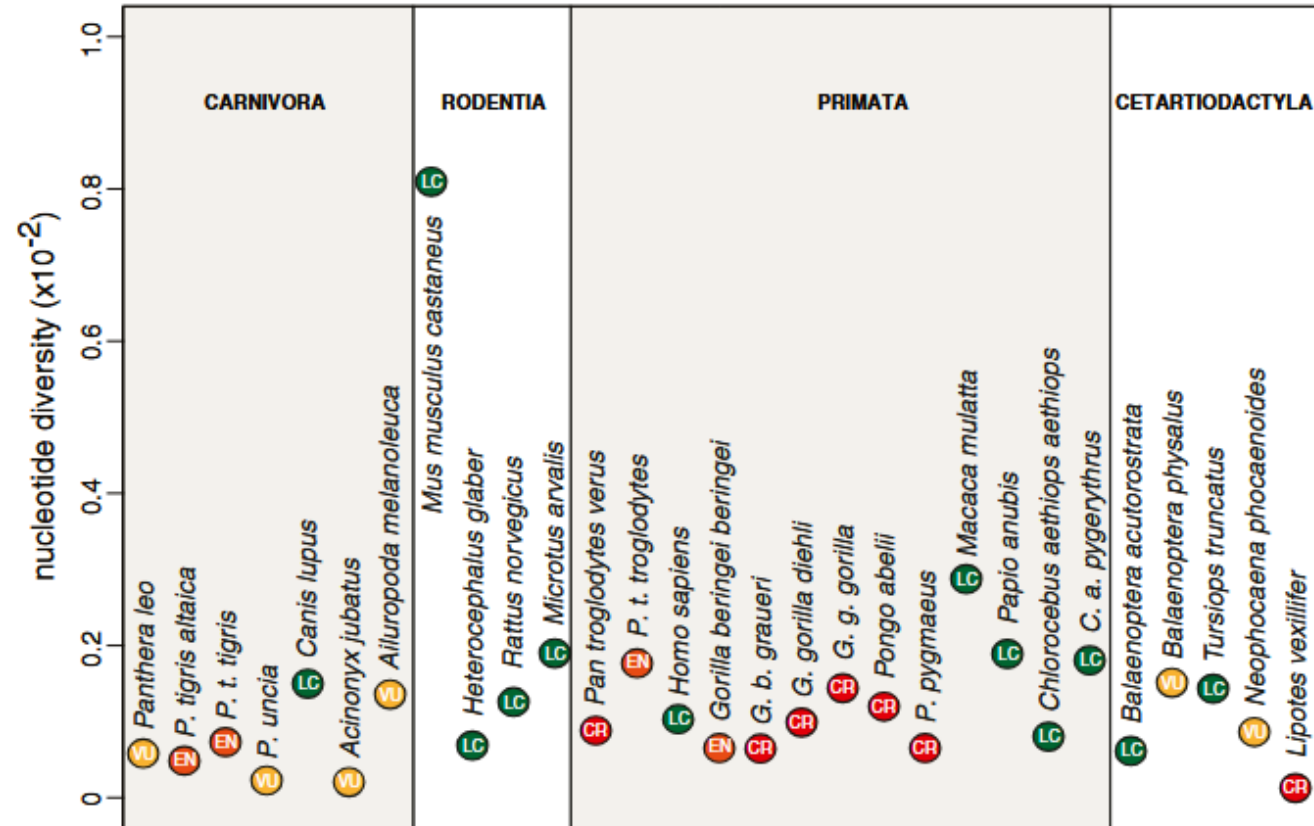
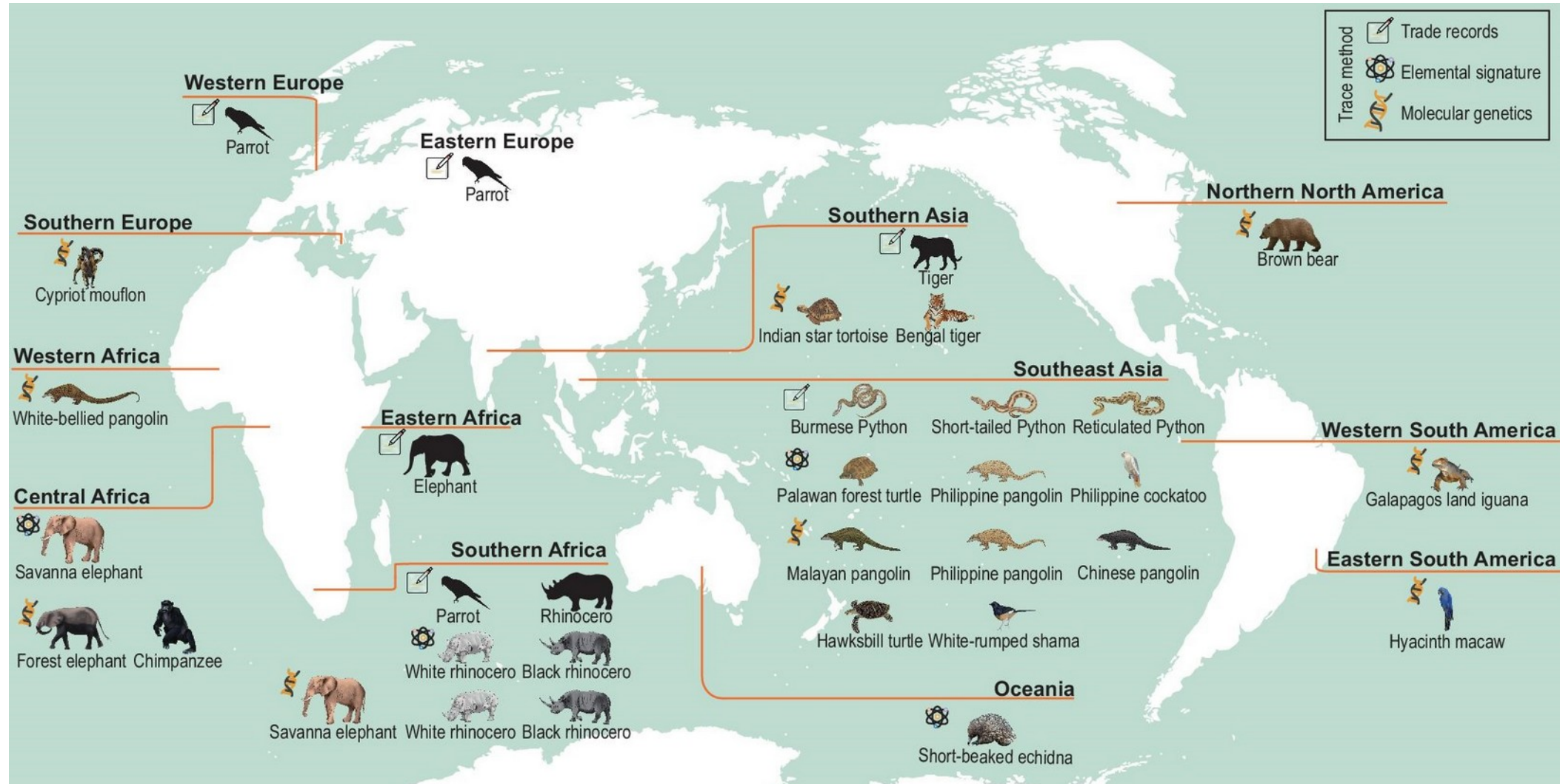


Fig. 1. Genome-wide nucleotide diversity is a poor predictor of IUCN's Red List status. Mammalian nucleotide diversity estimates were taken from Robinson et al. (42). Some critically endangered (CR) species, such as the Yangtze river dolphin or baiji (*Lipotes vexillifer*), show extremely low levels of nucleotide diversity. However, low levels of diversity are not necessarily due to recent anthropogenic pressures [the observed low genetic diversity in the baiji is due to a population bottleneck during the last glacial maximum ~20 kya (114)]. Other endangered (EN) and critically endangered species, such as the Sumatran orangutan (*Pongo abelii*), show levels of diversity that fall well within the range of least concern (LC) species.

CONSERVATION GENETICS AGAINST THE ILLEGAL TRADE OF ANIMALS

Natl Sci Rev, Volume 11, Issue 9, September 2024, nwae286, <https://doi.org/10.1093/nsr/nwae286>



WHAT IS THE PURPOSE OF POPULATION GENETICS

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Evolutionary parameters

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Conservation
genetics

Evolutionary
medicine

(PUT YOUR
OWNS)

“Molecular” processes

Stochastic
Evolutionary
process

Stochastic
Sampling
process

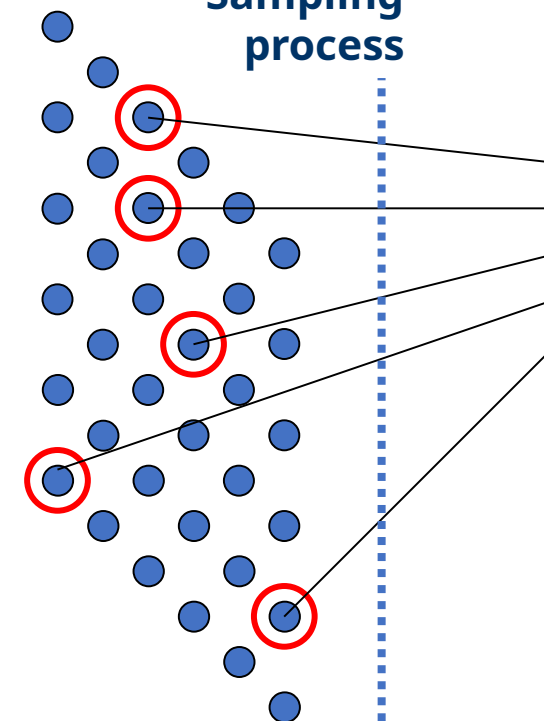
Mutation

Recombination

Selection

Genetic drift

Migration/Isolation



ATGCATGGGCTATTGGACCT
ATG**G**ATGGGCTATTG**C**ACCT
ATGCATGGGC**A**ATTG**C**ACCT
ATGCATGGGC**A**ATTGGACCT
ATG**G**ATGGGCTATTG**C**ACCT

Inference

Adapted from McVean

PHENOTYPE = GENOTYPE + ENVIRONMENT

Genotype

AA
GTGGT
ATTCT
TTCCA
TTCT
AATCTC
GCGGCG
TTTCTCA
AAAATC
TTTTGOGC
TCGCGGGCTA
TTGCTCAATCT
ATCTGCTC
TTAAGCTTC
TTCA TTAC
CGC TCGT
AATC AGGT
CATC CTCT
GGTA GGG
AAAT TTTCTA
CAGTCT TTC

Variability at a genomic level:
1 difference each ~1000
nucleotides
Small deletions-insertions
...

Phenotype



Each individual is unique

Environment

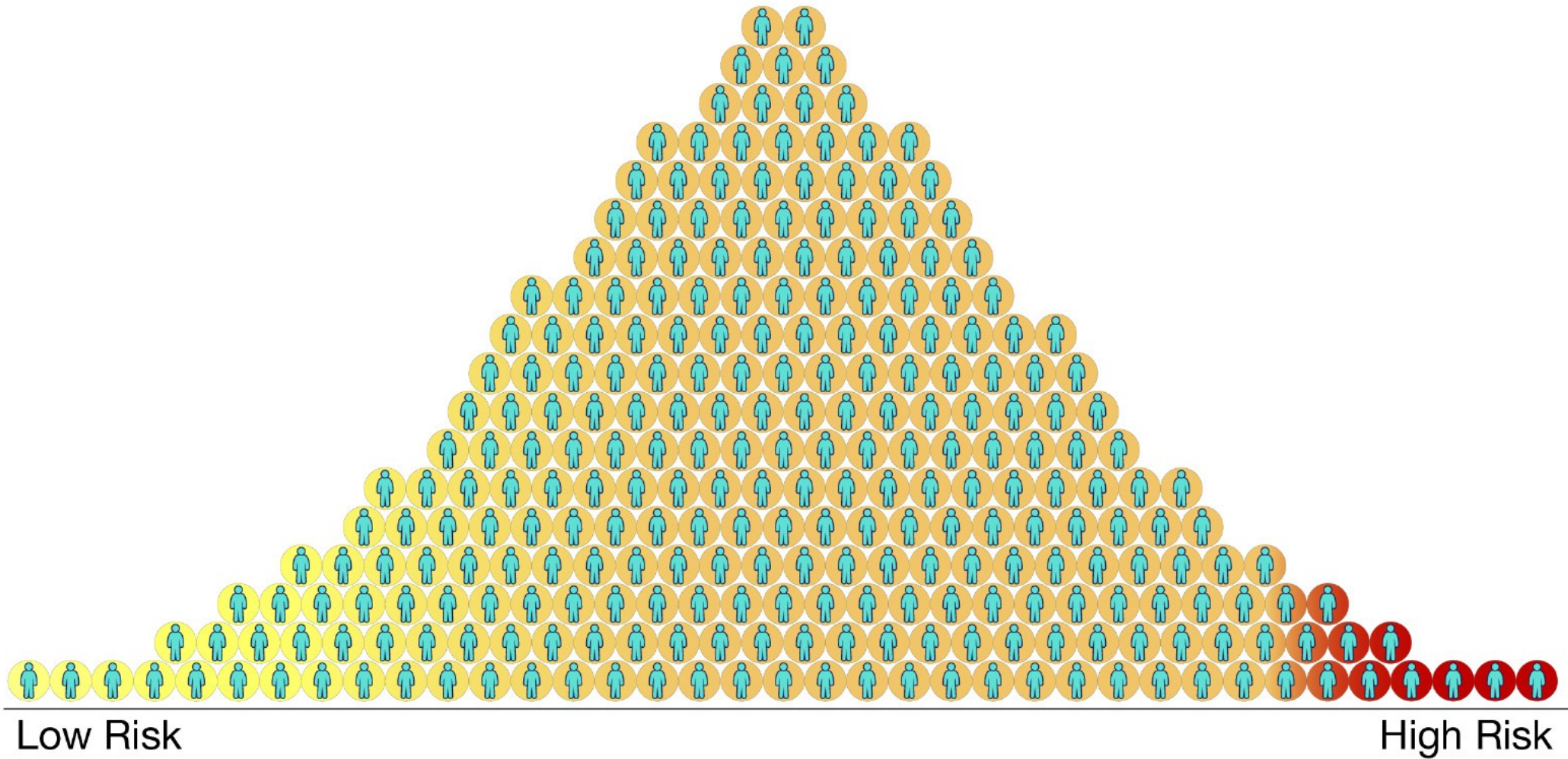


(... choose your environmental factor)

Variability at environmental
exposure level: particular
history of each individual

HOW TO IDENTIFY THE GENETIC CAUSE(S) OF A COMPLEX PHENOTYPE?

CURRENTLY ACCEPTED MODEL

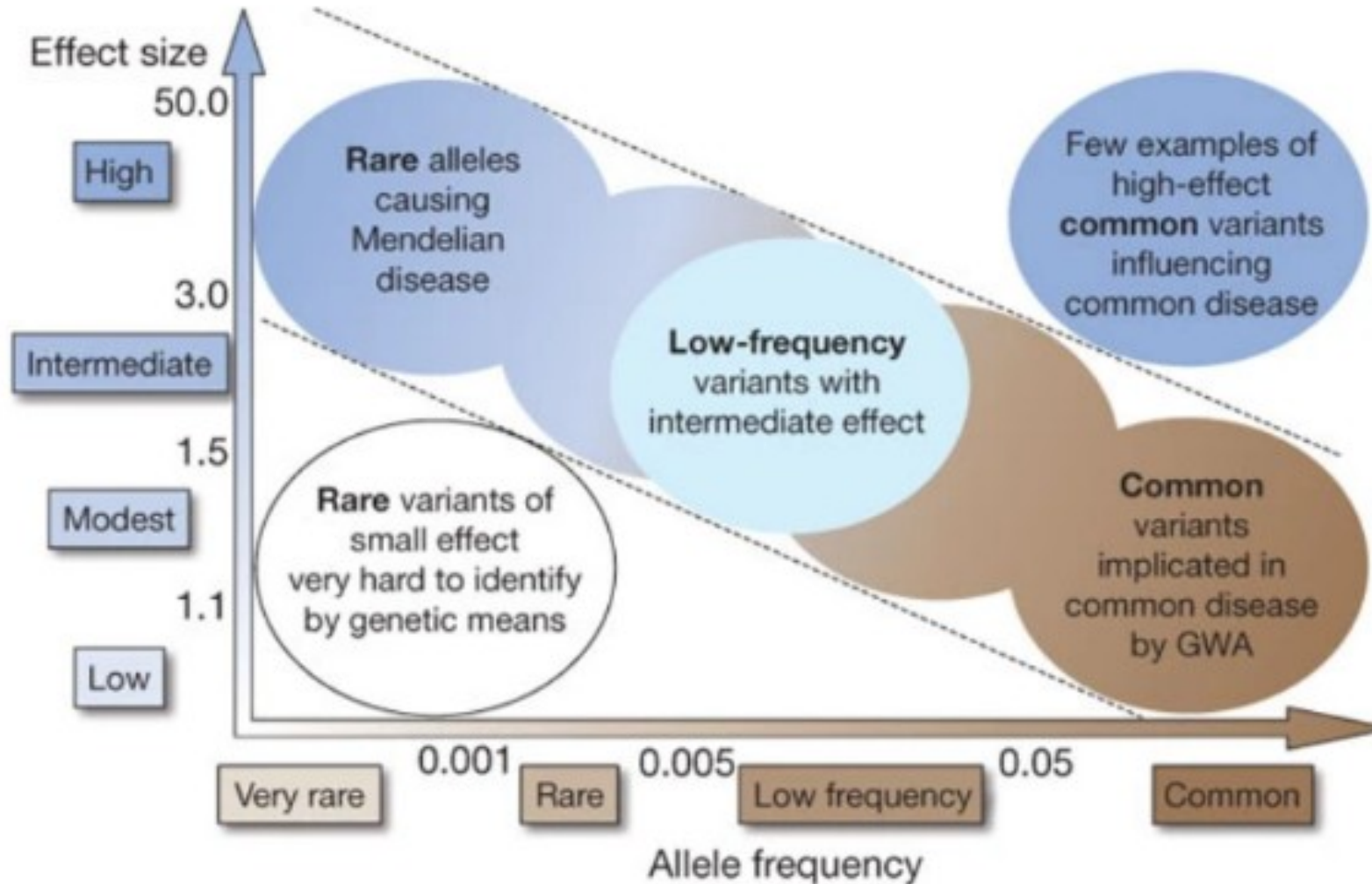


<https://www.genome.gov/Health/Genomics-and-Medicine/Polygenic-risk-scores>

HOW TO IDENTIFY THE GENETIC CAUSE(S) OF A COMPLEX PHENOTYPE?

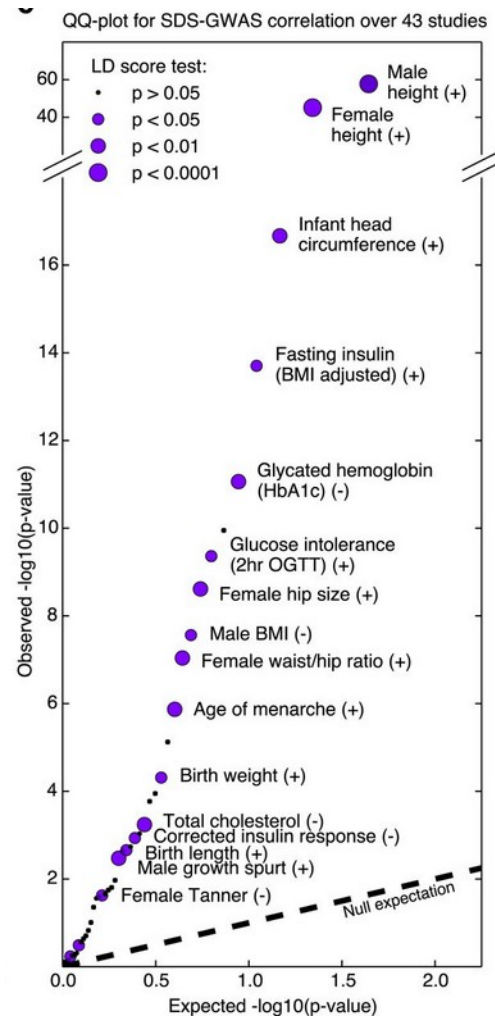
Finding the missing heritability of complex diseases.

Teri A. Manolio et al

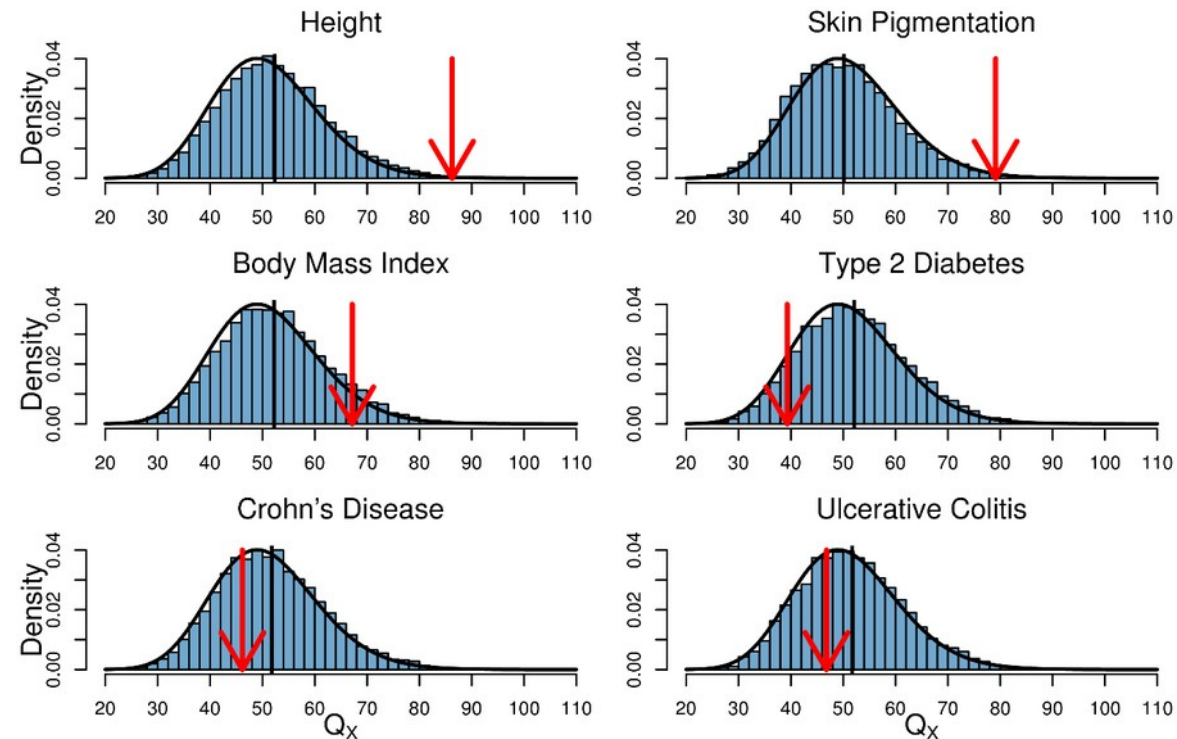


NOT ONLY MENDELIAN TRAITS ARE UNDER POSITIVE SELECTION

Field *et al.*, 2016. *Science*



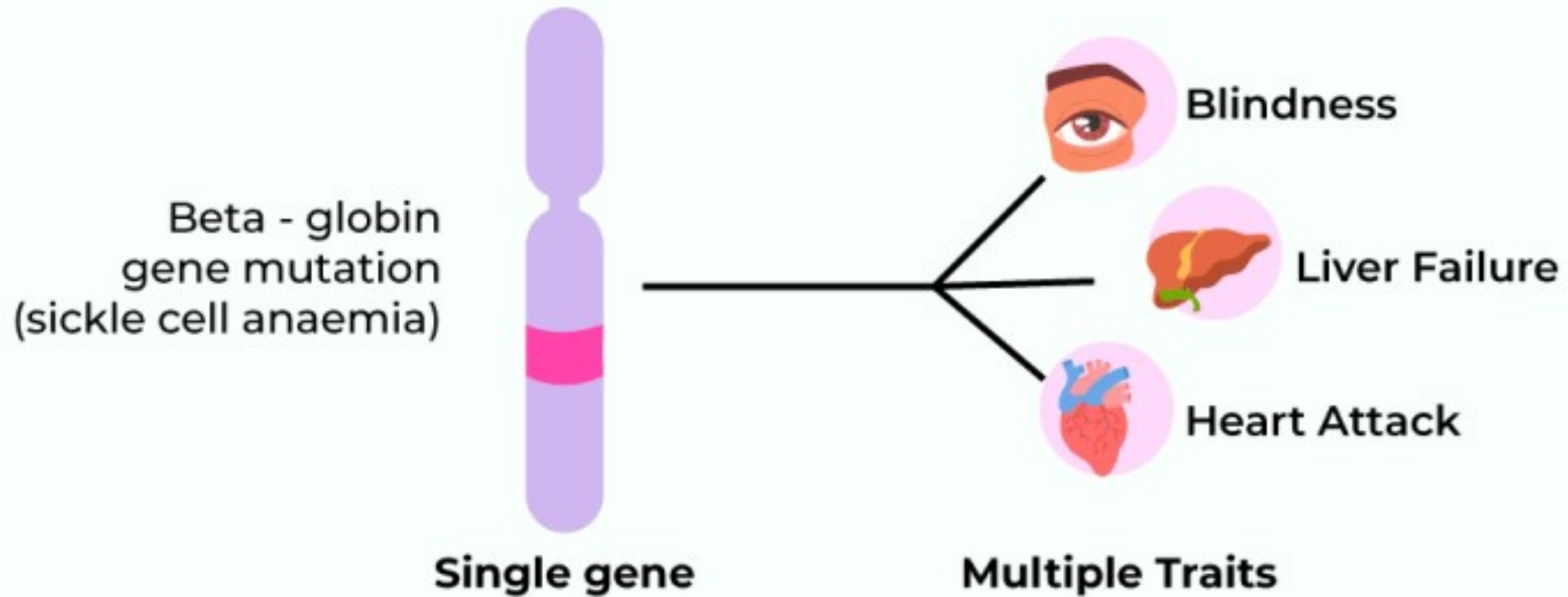
Berg and Coop., 2014. *Plos Genetics*



A MEDICAL CONSEQUENCE OF EVOLUTION

“After reproduction age, there is no need to maintain the integrity of the organism”

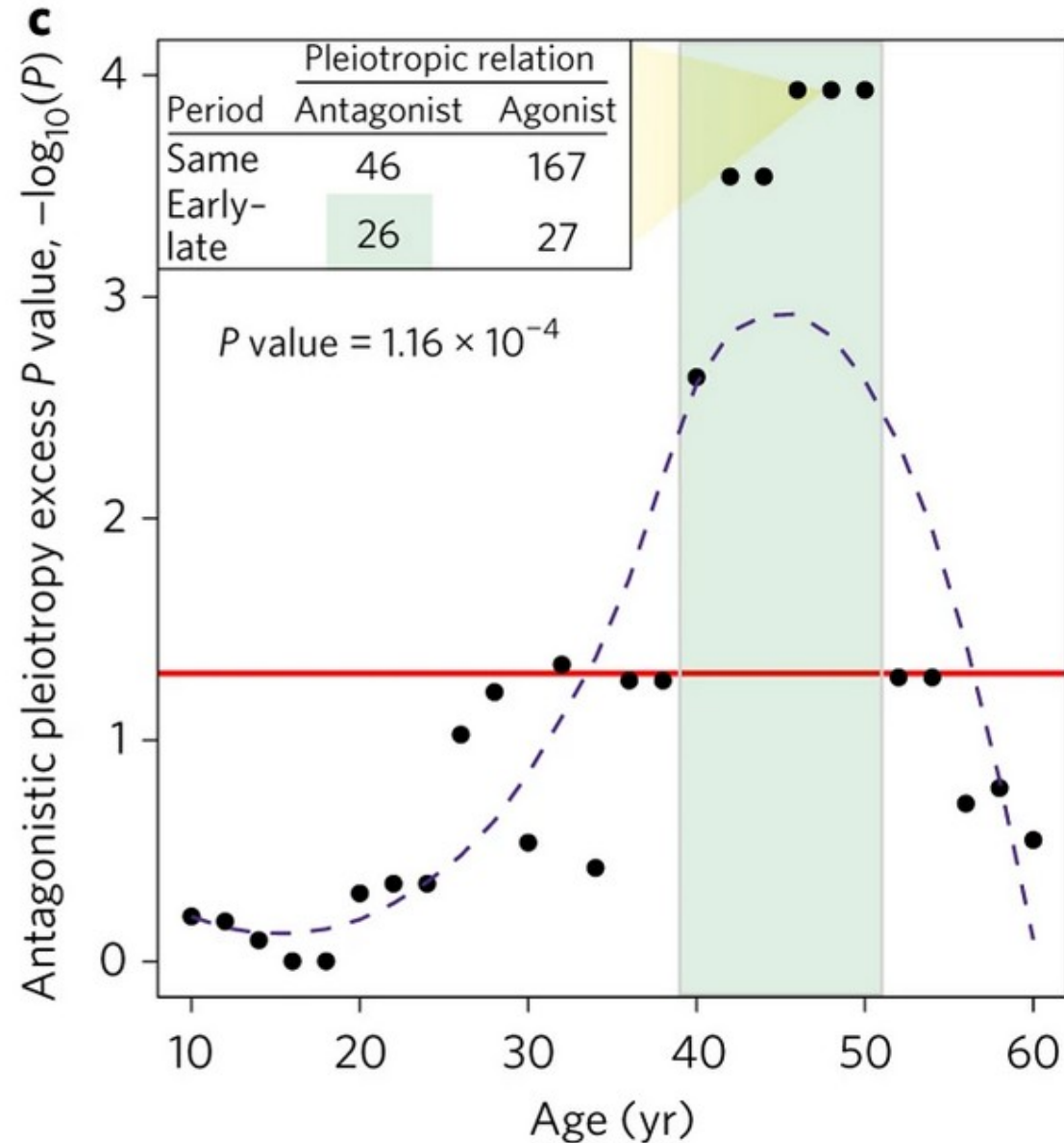
Pleiotropy



A MEDICAL CONSEQUENCE OF EVOLUTION

“After reproduction age, there is no need to maintain the integrity of the organism”

Rodriguez et al 2017

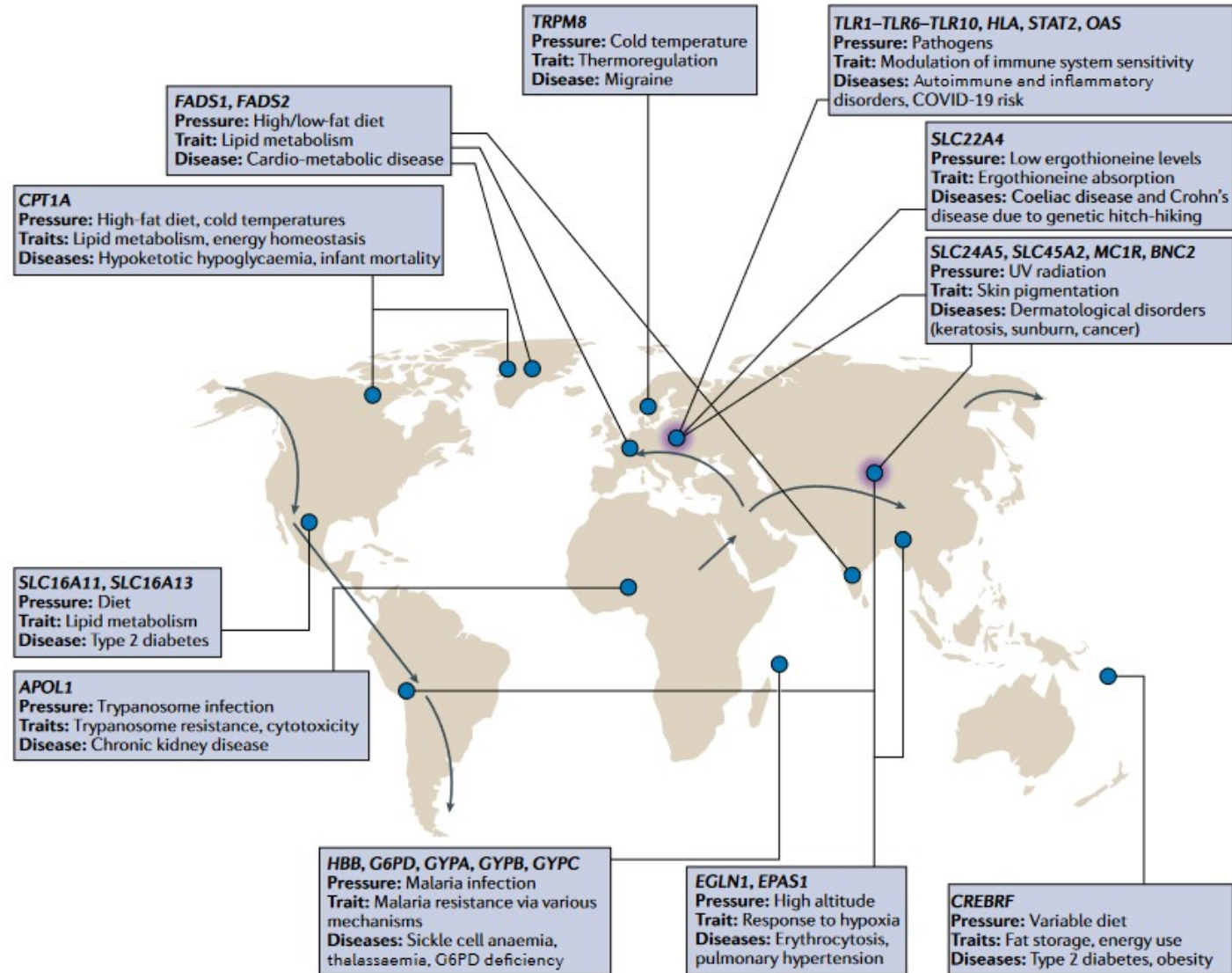


A MEDICAL CONSEQUENCE OF EVOLUTION

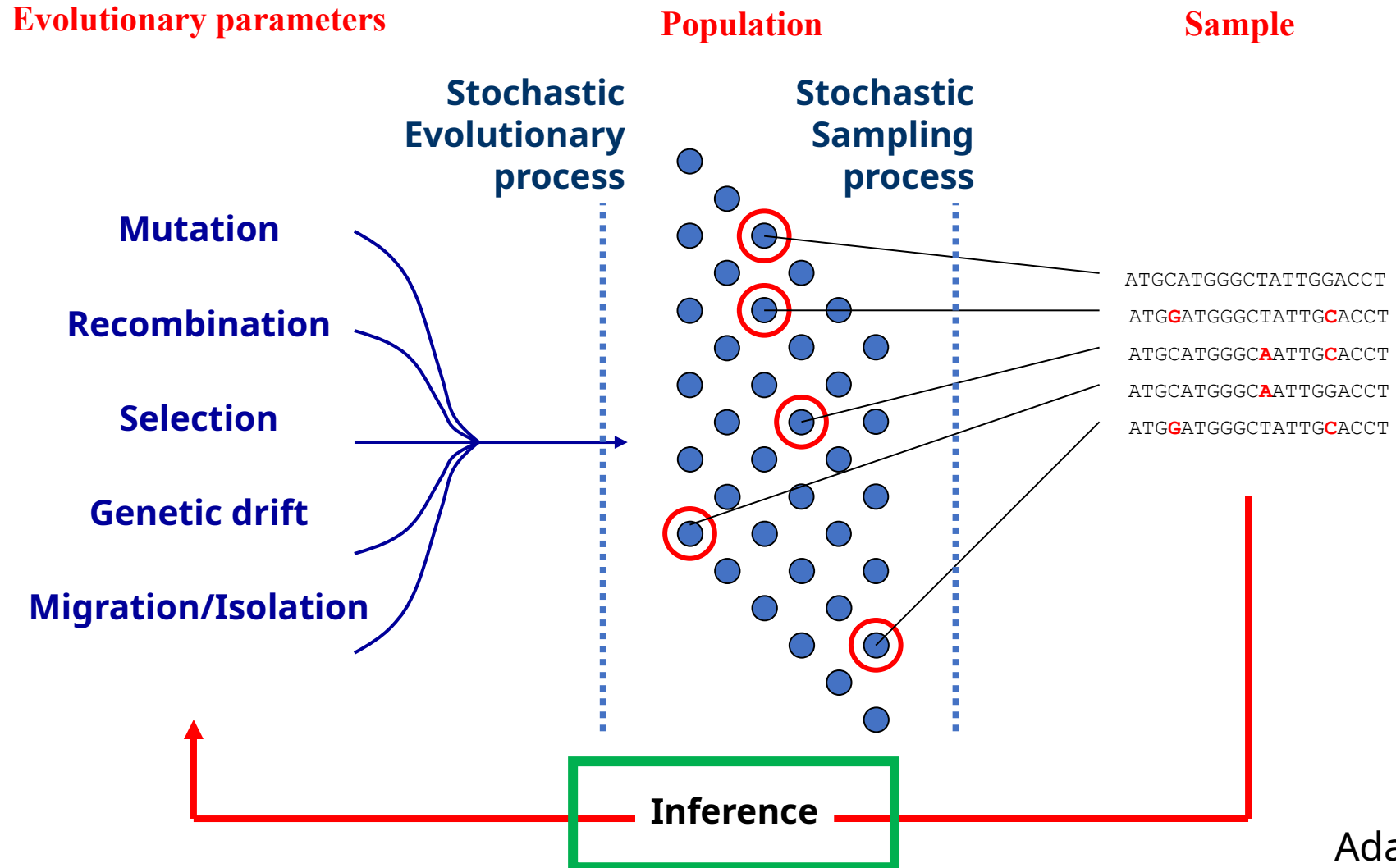
“After reproduction age, there is no need to maintain the integrity of the organism”

The influence of evolutionary history on human health and disease

Mary Lauren Benton, Abin Abraham, Abigail L. LaBella, Patrick Abbot, Antonis Rokas, and John A. Capra



WHAT IS THE PURPOSE OF POPULATION GENETICS



Properties of a Useful Summary Statistic in Population Genetics

- Informative: Captures relevant info about genetic structure and evolution
- Sensitivity to Population Parameters: Sensitive to mutation, selection, migration, etc.
- Low Bias: Provides unbiased estimates of population parameters
- Efficiency: Accurate estimates with minimal data
- Computationally Feasible: Can be calculated efficiently for large datasets
- Robustness: Works well with data deviations and model violations
- Interpretability: Clear biological/evolutionary interpretations
- Power: Ability to detect signals when they exist
- Ability to Integrate Across Loci: Aggregates info from multiple loci
- Appropriateness for the Model/Data Type: Suitable for specific evolutionary models
- Scalability: Reliable estimates as sample size increases
- Resilience to Population History Complexity: Detects different evolutionary scenarios

INFERENCE: WE NEED STATISTICS CAPTURING DIVERSITY!

Statistic	Description	Reference
Nucleotide Diversity (π)	Measures the average number of nucleotide differences per site between two randomly chosen individuals.	Nei, M. (1987). "Molecular Evolutionary Genetics." Columbia University Press.
Heterozygosity (H)	Measures the proportion of heterozygous loci in a population (observed or expected).	Nei, M. (1973). "Analysis of gene diversity in subdivided populations." Proceedings of the National Academy of Sciences.
Tajima's D	Compares nucleotide diversity (π) and the number of segregating sites (S), used to detect deviations from neutral evolution.	Tajima, F. (1989). "Statistical method for testing the neutral mutation hypothesis by DNA polymorphism." Genetics.
FST (Fixation Index)	Measures genetic differentiation between populations by comparing allele frequencies within and between populations.	Wright, S. (1951). "The genetical structure of populations." Annals of Eugenics.
Watterson's Theta (θ_W)	Estimates population mutation rate based on the number of segregating sites (S).	Watterson, G. A. (1975). "On the number of segregating sites in the model of neutral evolution." Theoretical Population Biology.
Site Frequency Spectrum (SFS)	Describes the distribution of allele frequencies in a sample, showing low, intermediate, and high frequencies.	Wakeley, J. (2009). "Coalescent Theory: An Introduction." Roberts and Company Publishers.
Allelic Richness	Measures the total number of different alleles at a locus across a population.	El Mousadik, A., & Petit, R. J. (1996). "High level of genetic differentiation for allelic richness among populations of the argan tree." Heredity.
Average Number of Alleles (N_a)	The average number of alleles observed per locus in the population.	Nei, M., & Kumar, S. (2000). "Molecular Evolution and Phylogenetics." Oxford University Press.
Number of Segregating Sites (S)	The number of nucleotide sites that are polymorphic in the sample.	Watterson, G. A. (1975). "On the number of segregating sites in the model of neutral evolution." Theoretical Population Biology.
Pi (π) vs Theta (θ) Comparison	Comparing nucleotide diversity (π) with Watterson's Theta (θ_W) to infer selection or demographic events.	Tajima, F. (1989). "Statistical method for testing the neutral mutation hypothesis by DNA polymorphism." Genetics.
Linkage Disequilibrium (LD)	Measures non-random association of alleles at different loci, indicating recombination or selection pressures.	Lewontin, R. C. (1964). "The interaction of selection and linkage. I. General considerations; heterotic models." Genetics.
Harvard's F (FIS)	Quantifies the inbreeding coefficient (degree of homozygosity within individuals relative to the population).	Wright, S. (1965). "The interpretation of population structure by F-statistics with special regard to systems of mating." Evolution.
ROH	Runs of homozygosity. Measures the fragments inherited from a recent ancestor. Indicative of inbreeding	McQuillan, R., et al. (2008). "Runs of Homozygosity in European Populations." American Journal of Human Genetics, 83(3), 359-372. https://doi.org/10.1016/j.ajhg.2008.08.007
Number of Private Alleles	The number of alleles that are unique to a particular population and not shared with others.	Nei, M. (1987). "Molecular Evolutionary Genetics." Columbia University Press.
Fay and Wu's H	Detects deviations from neutral evolution, specifically identifying selective sweeps in a genomic region.	Fay, J. C., & Wu, C. I. (2000). " hitchhiking under positive Darwinian selection." Genetics.
Ewens-Watterson Test	Compares observed allele frequencies to those expected under a neutral model, used to detect selection or demographic events.	Ewens, W. J., & Watterson, G. A. (1974). "The Ewens-Watterson test of neutrality." Theoretical Population Biology.
Substitution Rate	The rate at which mutations (substitutions) accumulate at a specific locus over time.	Kimura, M. (1983). "The Neutral Theory of Molecular Evolution." Cambridge University Press.
R2 Statistic	Measures the correlation (LD) between two loci, providing insight into the	Lewontin, R. C. (1964). "The interaction of selection and linkage. I. General

INFERENCE: WE NEED STATISTICS CAPTURING DIVERSITY!

Site Frequency Spectrum

Individuals

N

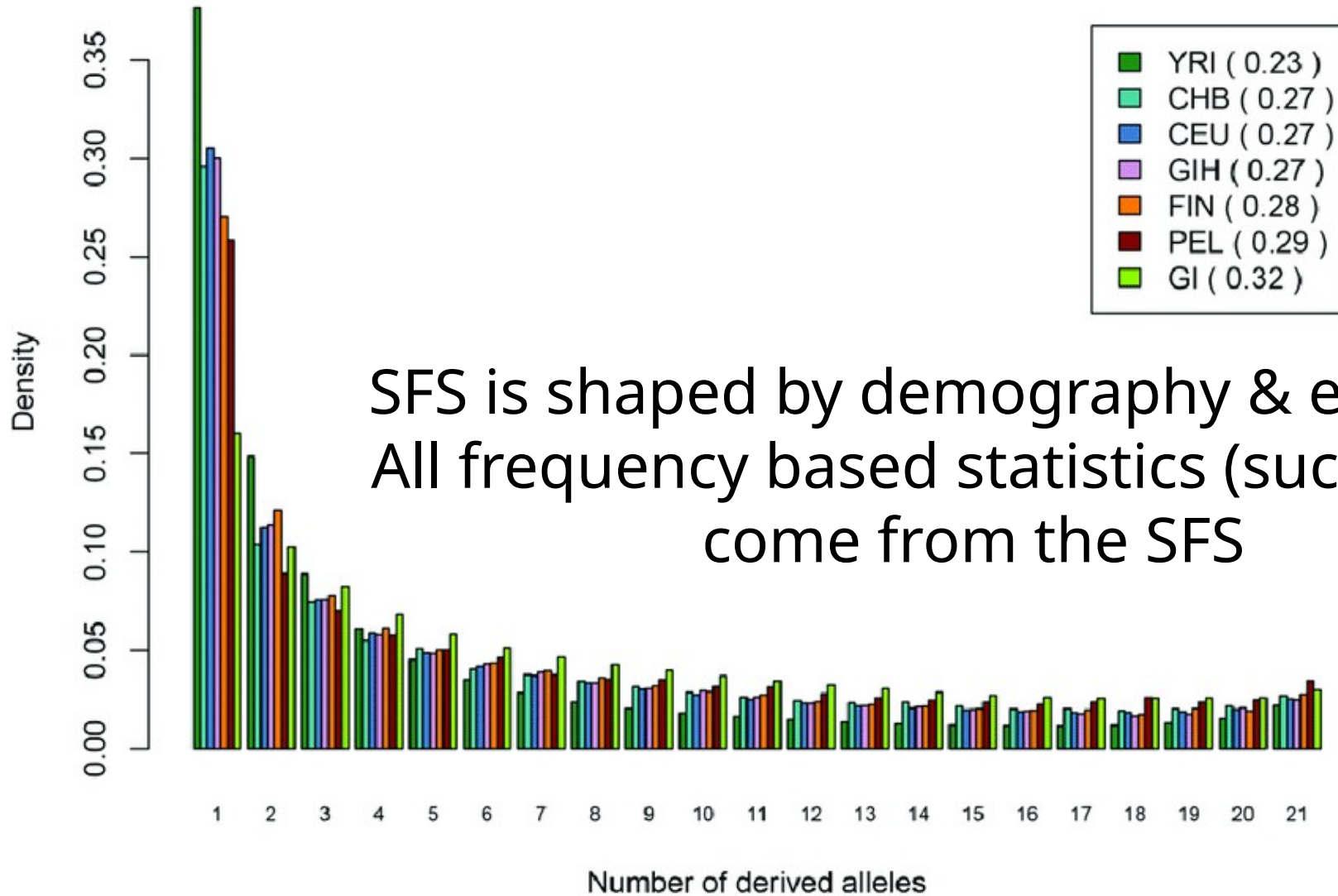
SNP1	SNP2	SNP3	SNP4	SNP5	SNP6	SNP7	SNP8	SNP9	SNP10
0	1	2	0	0	2	0	0	1	1
1	1	1	1	0	0	0	1	2	0
0	0	0	0	2	2	0	0	1	0
0	0	0	1	2	1	1	0	0	1
1	2	Number of SNPs with k derived alleles			Frequency			4	2

Genotypes
(number
of copies
of the
derived
allele)

1	3
2	2
3	1
4	2
5	1
6	0
7	0
8	0
9	0

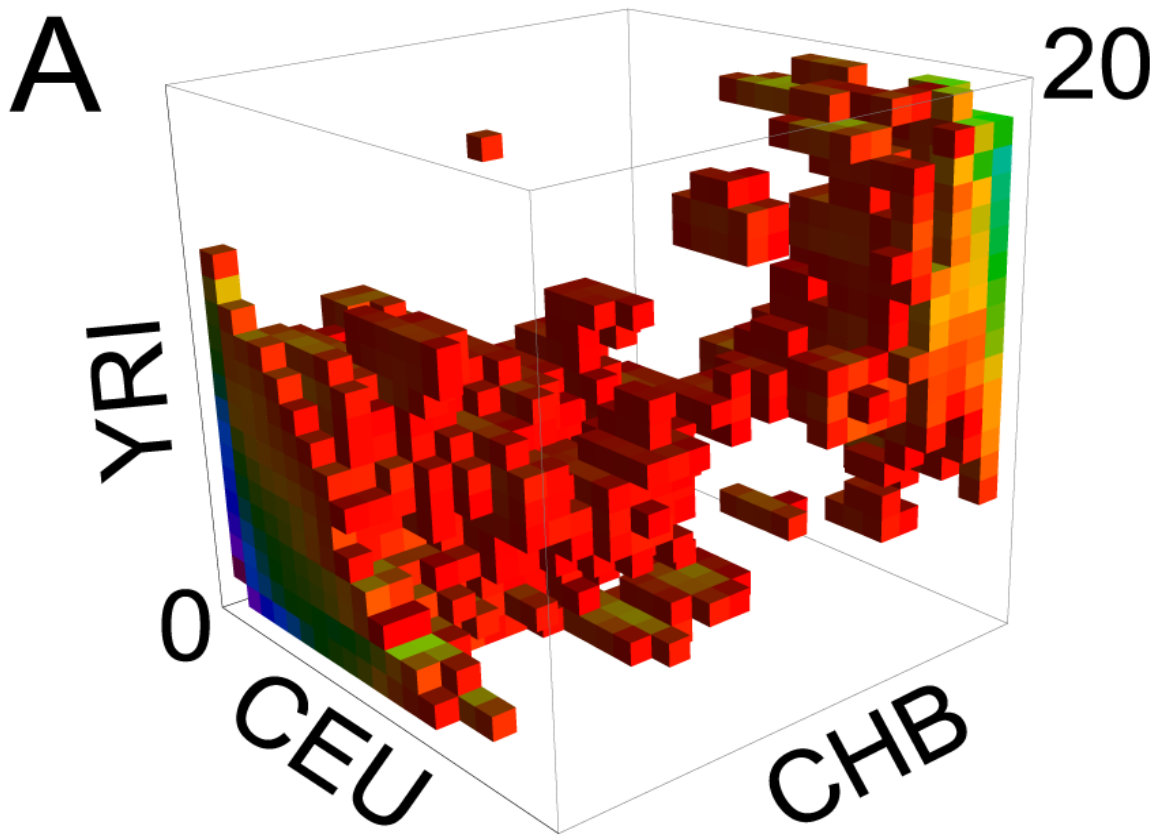
WHY SFS?

SFS comparison for six human populations



WHY SFS?

Unfolded Multidimensional Site Frequency Spectrum



Many statistics are derived from the SFS. -> SUPER INFORMATIVE

Problem: Curse of dimensionality

CURSE OF DIMENSIONALITY



As the quantity of features grows, patterns become less distinct, leading to a scenario where each data point possesses its own distinct characteristics.



INFERENCE: WE NEED STATISTICS CAPTURING POPULATION SUBSTRUCTURE!

Statistic/Method	Purpose	Key References
FST, FIS, FIT	Genetic differentiation & inbreeding	Wright (1951); Weir & Cockerham (1984)
HO, HE	Observed & expected heterozygosity	Nei (1973)
ORCA, Informativeness of ancestry	Observed & expected heterozygosity	Rosenberg (2003)
Genetic Distances	Nei's, Reynolds', Cavalli-Sforza distances	Nei (1972); Cavalli-Sforza & Edwards (1967)
PCA / DAPC	Multivariate clustering & structure visualization	Patterson et al. (2006); Jombart et al. (2010)
MAF & Allele Sharing	Frequency divergence; private/shared alleles	Nielsen et al. (2009); Excoffier & Slatkin (1995)
AMOVA	Partitioning variance within and among populations	Excoffier et al. (1992)
LD Decay	Historical recombination and structure	Slatkin (2008)
STRUCTURE / ADMIXTURE	Ancestry proportions & clustering	Pritchard et al. (2000); Alexander et al. (2009)
ROH / IBD	Inbreeding, recent shared ancestry	McQuillan et al. (2008); Browning (2010)
f3 / f4 Statistics	Testing for admixture and shared drift	Reich et al. (2009); Patterson et al. (2012)
Nucleotide Diversity (π)	Measures sequence-level diversity	Nei & Li (1979)

INFERENCE: WE NEED STATISTICS CAPTURING POPULATION SUBSTRUCTURE!

- F_{st} based on frequency

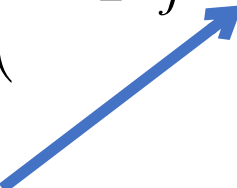
$$F_{st} = \frac{\frac{(p_a - \hat{p})^2 + (p_b - \hat{p})^2}{2}}{2\hat{p}(1 - \hat{p})}$$

$$\hat{p} = \frac{p_a + p_b}{2}$$

INFERENCE: WE NEED STATISTICS CAPTURING POPULATION SUBSTRUCTURE!

- Informativeness of ancestry

$$I_n(Q; J) = \sum_{j=1}^N \left(- p_j \log p_j + \sum_{i=1}^K \frac{p_{ij}}{K} \log p_{ij} \right)$$



**Entropy of the mean
allelic distribution
(that, is we assume
that all the
individuals are from
the same population)**



**Mean entropy of the
allelic distribution in all
the populations**

INFERENCE: WE NEED STATISTICS CAPTURING THE FINGERPRINT OF SELECTION!

Statistic	Description	Reference
Tajima's D	Compares the average number of pairwise differences (π) and the number of segregating sites (S) to test for deviations from neutral evolution. Positive values suggest balancing selection; negative values suggest a selective sweep or population expansion.	Tajima, F. (1989). "Statistical method for testing the neutral mutation hypothesis by DNA polymorphism." Genetics.
Fay and Wu's H	Detects recent positive selection by comparing the observed distribution of allele frequencies to that expected under neutrality. Negative values indicate a recent selective sweep.	Fay, J. C., & Wu, C. I. (2000). "Hitchhiking under positive Darwinian selection." Genetics.
McDonald-Kreitman Test	Compares the ratio of polymorphisms to fixed differences in both coding and non-coding regions to detect selective pressures acting on a gene. A significant departure from the neutral model suggests selection.	McDonald, J. H., & Kreitman, M. (1991). "Adaptive evolution at the Adh locus in Drosophila." Nature.
HKA Test (Hudson-Kreitman-Aguadé)	Tests whether variation within a population and between populations fits expectations under neutral evolution. A significant result indicates selection.	Hudson, R. R., Kreitman, M., & Aguadé, M. (1987). "A test of neutral molecular evolution based on nucleotide data." Genetics.
Ewens-Watterson Test	Compares observed allele frequencies to expected frequencies under the neutral model. Significant deviations indicate non-neutral evolution (e.g., selection).	Ewens, W. J., & Watterson, G. A. (1974). "The Ewens-Watterson test of neutrality." Theoretical Population Biology.
Linkage Disequilibrium (LD)	Measures the non-random association of alleles at different loci. High levels of LD may suggest selective sweeps or genetic hitchhiking, while low LD suggests recombination.	Lewontin, R. C. (1964). "The interaction of selection and linkage. I. General considerations; heterotic models." Genetics.
D/Fay's H	A variant of Fay and Wu's H used to detect selection by evaluating the number of derived alleles compared to the expected number under neutrality.	Fay, J. C., & Wu, C. I. (2000). "Hitchhiking under positive Darwinian selection." Genetics.
EHH (Extended Haplotypic Homogeneity)	Measures the extent of homozygosity across large regions of the genome to identify regions of selective sweeps. Low EHH at a locus may indicate selection.	Sabeti, P. C., et al. (2002). "Positive natural selection in the human genome." Science.
iHS (Integrated Haplotype Score)	Used to detect selective sweeps by comparing haplotype lengths across a region of the genome. It focuses on recent positive selection, with low iHS values indicating a sweep.	Voight, B. F., et al. (2006). "A map of recent positive selection in the human genome." PLOS Biology.
nSL (Normalized Selection Score)	Measures the length of extended haplotypes to identify evidence of recent positive selection. It is useful for identifying regions of the genome under strong selective pressure.	Felsenstein, J., & Yokoyama, S. (2010). "Quantifying natural selection by the use of an integrated haplotype score." PLOS Genetics.
Pernicious (Locus-specific Perplexity Test)	A statistical test for detecting deviations from neutrality, with a focus on loci with complex selection signatures. It tests for shifts in allele frequencies indicating selective pressures.	Messer, P. W., & Petrov, D. A. (2013). "Pernicious: a test for local adaptation based on population genomics." Evolutionary Biology.
Skelly's Test	A method used for detecting selective sweeps based on the observed allele frequency spectrum. A significant result suggests positive selection.	Skelly, G. P., & Roberts, J. R. (1998). "Detecting selective sweeps in population genetics." Genetics.

INFERENCE: WE NEED STATISTICS USING LINKAGE DISEQUILIBRIUM!

Statistic	Description	Reference
D' (D-prime)	A measure of the degree of linkage disequilibrium between two loci, ranging from -1 to +1. A value of 1 indicates complete association between alleles.	Lewontin, R. C. (1964). "The interaction of selection and linkage. I. General considerations; heterotic models." Genetics.
r ² (Correlation Coefficient)	Measures the correlation between alleles at two loci. The value of r ² ranges from 0 (no association) to 1 (perfect association).	Hill, W. G., & Robertson, A. (1968). "Linkage disequilibrium in finite populations." Theoretical and Applied Genetics.
Hill-Robertson Effect	Describes the reduction in genetic diversity due to the effect of selection on linked loci. It can be quantified by comparing expected vs observed LD.	Hill, W. G., & Robertson, A. (1968). "Linkage disequilibrium in finite populations." Theoretical and Applied Genetics.
FST (Fixation Index)	Measures genetic differentiation between populations. LD is often used to examine FST between loci within a population or across populations.	Wright, S. (1951). "The genetical structure of populations." Annals of Eugenics.
Lewontin's D	A measure of the deviation from expected allele frequencies under random mating. It ranges from -1 to +1, with 0 indicating no linkage.	Lewontin, R. C. (1964). "The interaction of selection and linkage. I. General considerations; heterotic models." Genetics.
Haplotype Diversity	Measures the diversity of haplotypes (combinations of alleles at different loci) in a population. Low haplotype diversity suggests high LD.	Thompson, E. A. (2000). "Haplotype diversity." Genetics.
Extended Haplotype Homozygosity (EHH)	Measures the extent of LD around a locus. Extended EHH is often used to detect selective sweeps by looking for long haplotypes in populations.	Sabeti, P. C., et al. (2002). "Positive natural selection in the human genome." Science.
Integrated Haplotype Score (iHS)	A statistic that detects regions of the genome under recent positive selection by comparing haplotype structures and linkage disequilibrium.	Voight, B. F., et al. (2006). "A map of recent positive selection in the human genome." PLOS Biology.
D (Delta-prime)*	A modification of D' that accounts for the presence of rare alleles in a population. It is particularly useful in populations with low minor allele frequencies.	Stephens, M., & Donnelly, P. (2003). "A comparison of Bayesian methods for haplotype reconstruction." American Journal of Human Genetics.
r ² (Pairwise LD Coefficient)	A pairwise measure of LD between two loci based on allele frequencies. It is widely used in genome-wide association studies (GWAS).	Weir, B. S. (1996). "Genetic data analysis II: Methods for discrete population genetic data." Sinauer Associates.
LD Score Regression (LDSC)	A method for estimating the heritability of traits from LD scores. It regresses traits on the LD score, allowing the detection of associations between traits and regions of high LD.	Bulik-Sullivan, B., et al. (2015). "LD Score regression distinguishes confounding from polygenicity in genome-wide association studies." Nature Genetics.
Two-Locus LD	Measures the degree of association between alleles at two loci. Commonly used to calculate both D' and r ² for pairwise loci in	Nei, M., & Roychoudhury, A. K. (1974). "Sampling variances of heterozygosity and genetic distances." Genetics.

INFERENCE: WE NEED STATISTICS CAPTURING ARCHAIC INTROGRESSION!

Statistic	Description	Reference
FST (Fixation Index)	FST can be used to detect genomic regions that show high differentiation between modern human populations and archaic populations, indicating introgression.	Wright, S. (1951). "The genetical structure of populations." Annals of Eugenics.
Introgression Detection (iHS)	Measures the integration of archaic alleles by analyzing extended haplotype homozygosity (EHH). Archaic alleles under introgression are detected by long haplotypes.	Sabeti, P. C., et al. (2002). "Positive natural selection in the human genome." Science.
Admixture Mapping	A method used to identify regions of the genome that have been introgressed from an ancestral population (e.g., Neanderthal or Denisovan). It compares allele frequencies across populations.	Pritchard, J. K., et al. (2010). "Incorporating heterogeneity into the model for local ancestry inference." The American Journal of Human Genetics.
ABBA-BABA Test	A test that detects introgression by comparing allele frequencies between two populations (archaic and modern). Used to test for gene flow between populations.	Green, R. E., et al. (2010). "A draft sequence of the Neanderthal genome." Science.
D-statistic (ABBA-BABA test)	The D-statistic compares allele frequencies between a group of modern human populations and archaic populations (e.g., Neanderthals, Denisovans). Positive values indicate introgression.	Patterson, N., et al. (2012). "Ancient admixture in human history." Nature.
Haplotype-based Tests	Tests that focus on long haplotypes in specific regions, which are signatures of archaic introgression. These can be detected by comparing populations with and without introgressed haplotypes.	Vernot, B., & Akey, J. M. (2014). "Resurrecting surviving Neandertal lineages from modern human genomes." Science.
Identity-by-Descent (IBD)	Analyzes regions of the genome that are identical by descent, which can indicate recent introgression from archaic populations.	Slatkin, M. (2008). "Linkage disequilibrium—understanding the evolutionary forces shaping the genome." Nature Reviews Genetics.
Neanderthal-specific SNPs	Identifies single nucleotide polymorphisms (SNPs) that are unique to Neanderthals or Denisovans, and their presence in modern populations can be a signature of introgression.	Reich, D., et al. (2010). "Genetic history of an archaic hominin group from Denisova Cave in Siberia." Nature.
SNP Frequency Spectrum	A method used to compare the frequency of SNPs between archaic and modern populations. Differences in the frequency spectrum can suggest introgression.	Meyer, M., et al. (2012). "A high-coverage genome sequence from an archaic Denisovan individual." Science.
Local Ancestry Inference (LAI)	Local ancestry inference methods use genetic data to determine the proportion of an individual's genome derived from archaic populations (e.g., Neanderthals, Denisovans).	Henn, B. M., et al. (2012). "Population genetics of the Neolithic transition in Europe." PLOS Biology.
Mendelian Randomization (MR)	Uses genetic variants that have been introgressed from archaic populations to investigate their effects on traits in modern human populations.	Burgess, S., & Thompson, S. G. (2015). "Mendelian Randomization: Methods for Using Genetic Variants in Causal Estimation." CRC Press.
Haplotype-based Phylogenetic Analysis	Phylogenetic methods analyze the structure of haplotypes to detect introgression from archaic populations. Such methods compare the genealogical relationships of alleles across populations.	Vernot, B., & Akey, J. M. (2014). "Resurrecting surviving Neandertal lineages from modern human genomes." Science.
Clade-specific Analysis	This method looks for specific genomic clades in modern human populations that are genetically similar to archaic populations, indicating gene flow.	Sankararaman, S., et al. (2012). "The distribution of Neanderthal ancestry in present-day humans." Nature.

Usage of summary statistics

Exploratory analyses. “Do my data look OK?” Sequencing errors, sampling errors, Simulation errors, ERRORS, ERRORS, ERRORS...

Data cleaning. For exemple, HWE and sequencing errors.

Evolutionary inferences: given an evolutionary model for which summmary statistic is targeted

Transfer learning. The statistic is created for another evolutionary model (pooling summary statistics can help) ABC

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PACKAGES IN R

Package	Main Uses	Key References
adegenet	Multivariate analysis (PCA, DAPC), genetic data handling	Jombart (2008), Jombart & Ahmed (2011)
hierfstat	F-statistics (Weir & Cockerham), AMOVA, population structure	Goudet (2005)
pegas	Haplotype networks, genetic diversity, neutrality tests	Paradis (2010)
vcfR	Read and process VCF files, integrate with adegenet	Knaus & Grünwald (2017)
poppr	Clonal population structure, genotypic diversity, multilocus data	Kamvar et al. (2014)
SNPRelate	Genome-wide SNP analysis, PCA, relatedness, IBD	Zheng et al. (2012)
LEA	Population structure inference, environmental association (LFMM)	Frichot & François (2015)
stAMPP	Pairwise FST, Nei's distances, AMOVA with stratified populations	Pembleton et al. (2013)
genepop	HWE, FST, isolation-by-distance, linkage disequilibrium	Raymond & Rousset (1995)
radiator	Filtering and format conversion for SNP datasets	Gosselin (2020)
ade4	Multivariate analyses, distance-based methods (supports adegenet)	Dray & Dufour (2007)
OutFLANK	Detecting loci under selection using FST outliers	Whitlock & Lotterhos (2015)

PACKAGES IN PYTHON

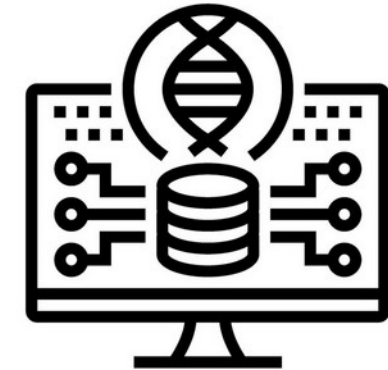
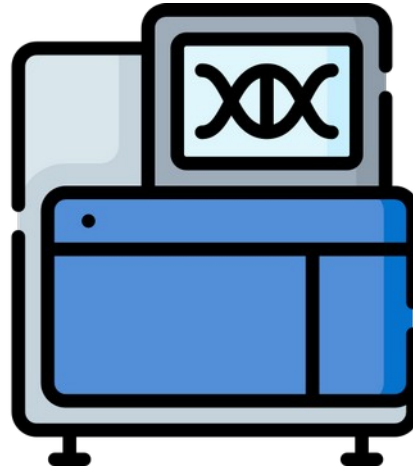
Package	Main Uses	Key References
scikit-allel	Analysis of VCF/genotype data, diversity, FST, PCA	Miles & Harding (2017)
dadi	Demographic inference from allele frequency spectra (AFS)	Gutenkunst et al. (2009)
msprime	Efficient coalescent simulations with recombination	Baumdicker et al. (2022)
tskit	Tree sequence analysis, storage, simulation output parsing	Kelleher et al. (2019)
moments	Extension of dadi for faster and flexible demographic modeling	Jouganous et al. (2017)
pyslim	Interface between SLiM and tskit tree sequences	Haller et al. (2019)
cyvcf2	Fast VCF parsing and filtering	Pedersen & Quinlan (2017)
popgenome	Population genetic summary statistics (via R; Python wrappers exist)	Pfeifer et al. (2014)
vcftools (CLI)	Not a Python package but often used with Python pipelines	Danecek et al. (2011)
gwaspy	GWAS support tools, PCA, quality control	Internal tools (community-contributed)
SLiM (via pyslim)	Forward-time simulations, evolutionary modeling	Haller & Messer (2019)
pyseer	Bacterial GWAS, fixed and mixed models	Lees et al. (2018)
snakemake	Workflow management for genomic pipelines	Mölder et al. (2021)

THE PAINFUL WAY OF SCIENCE

Hypothesis testing

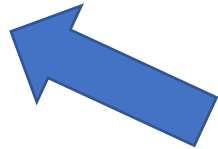
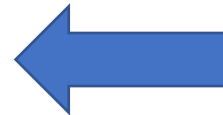
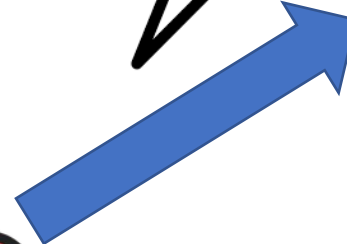
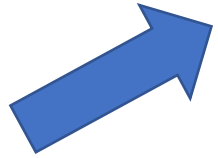
Sequencing

Data preprocessing



Interpretation of results

Genomic analyses



(SOME) BASIC STEPS OF DATA PREPROCESSING



Step	Purpose	Common Tools / Methods
1. Quality Assessment	Evaluate raw data quality (e.g., Phred scores, read length, GC content)	FastQC, MultiQC
2. Adapter & Quality Trimming	Remove low-quality bases and sequencing adapters	Trimmomatic, Cutadapt, fastp
3. Read Mapping / Alignment	Align reads to reference genome	BWA, Bowtie2, STAR, HISAT2
4. Duplicate Removal	Remove PCR duplicates that can bias variant calling	Picard (MarkDuplicates)
5. Sorting & Indexing	Organize BAM files for efficient access	samtools sort, samtools index
6. Base Quality Recalibration	Adjust quality scores to correct for systematic errors	GATK BaseRecalibrator
7. Variant Calling	Detect SNPs and indels from aligned reads	GATK HaplotypeCaller, FreeBayes
8. Variant Filtering	Remove low-confidence variants	GATK VariantFiltration, bcftools
9. Handling Missing Data	Impute or filter missing genotypes	BEAGLE, PLINK, VCFtools
10. Individual and SNP QC	Remove individuals or SNPs with high missingness or	PLINK, vcftools, plinkQC

BEWARE: GARBAGE IN GARBAGE OUT

- **Setting biased hypotheses and bad study designs**
- **Each step of preprocessing (for exemple, consider only markers above certain MAF, not removing LD...) can bias results.**
- **Merging datasets: batch effects, misleading the state of the alleles, different genomic builds, rsIds, etc.**
- **Performance of the algorithms depends on hyperparameters, which must be definined a priori. Default paràmetres maybe not the best for your data**
- **Nature of the summary statistics. Each makes its own assumptions (panmixia, constant population size...). Intepretation will be biased.**

Thank you very much!

